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The anatomy of behavioral responses to social assistance when informal employment is high ☆

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ABSTRACT

The disincentive effects of social assistance programs on registered (or formal) employment are a first-order policy concern in developing and middle-income countries. We study the impact of a conditional cash transfer (CCT) program in Uruguay on the employment of adult members in beneficiary households in a context of high informality. Our research design relies on the sharp discontinuity introduced by program eligibility rules around a poverty score threshold combined with longitudinal administrative data. We find reductions of about 6 percentage points (a 13% drop) in formal labor force participation among all beneficiaries and of 8.7 percentage points (a 19% drop) for single mothers. The implied elasticity of participation in the formal sector with respect to the net-of-tax rate is about 0.78 for the full sample and about 1.3 for single mothers. The reduction in labor supply is stronger among individuals who have a medium propensity to be formally employed, with a smaller reduction in the case of infra-marginal individuals. We also present suggestive evidence that the reduction in formal employment increases inactivity and informal work in equal proportions. Finally, despite pervasive informality in the context of the Family Allowance assistance program (AFAM), the program's marginal value of public funds of 0.61 implies an efficiency cost within the range of cash transfer programs targeted to families in the United States.

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1. Introduction

The incentive effects of social assistance programs on labor supply have been at the center of economic policy debate in developed countries. In developing and middle-income countries with high levels of labor informality and poor enforcement of tax and labor regulations, the difficulties of targeting have potential consequences beyond overall labor supply.[1] Social assistance programs may introduce disincentives to registered (formal) employment, which has important implications. First, access to social insurance for workers and their families is typically tied to formal jobs (Levy, 2008; Levy and Schady, 2013). Second, unregistered employment means lower reporting of earnings and thus lower payroll and

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[1] In this paper, "formal employees" refer to those workers who are compliant with social insurance regulations and do not evade payroll taxes. We use the terms "registered (unregistered)" and "formal (informal)" interchangeably, and both terms refer to the status of the worker with respect to the Social Security Administration.

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income-tax revenue. Third, a larger informal sector can lead to a variety of market distortions and efficiency losses, with potential consequences for productivity growth and economic development (Porta and Shleifer, 2014; Meghir et al., 2015). In Latin America, the recent expansion of conditional cash transfer (CCT) programs has prompted discussions of their effects on labor formality.[2] Empirical evidence on this specific issue, however, is limited and the impact of CCT programs on overall employment seems to be small or null (Alzua et al., 2012; Banerjee et al., 2015). Moreover, with few exceptions (Gerard and Gonzaga, forthcoming), these discussions are limited to labor market outcomes in isolation and failto quantify their welfare/efficiency costs.

Our analysis attempts to fill in some of these gaps in the existing literature by focusing on three main objectives.[3] We first establish the impact on formal employment of a CCT program –the Family Allowance (henceforth, AFAM) –with a binding-income threshold and frequent income reassessment in Uruguay, a country with a high level of labor informality. Moreover, labor market attachment varies substantially across beneficiaries, some of whom may not be at the margin of (in-) formality. The CCT program in Uruguay, like many others in the developing world, has a "one-size-fits-all" approach that fails to accommodate these features, with potentially negative consequences. Our second objective is, thus, to study the heterogeneity of worker responses in terms of attachment to the formal sector and to decompose the formal employment response into informality and non-employment margins. Our third objective is to evaluate the implications of our results in terms of efficiency costs. We use the responses we estimate to recover the marginal value of public funds (MVPF) following Hendren's (2016) methodology.

The empirical analysis relies on three matched sets of information linked through unique individual identifiers. The AFAM administrative records contain baseline socioeconomic and demographic information provided by applicants to the program from January 2008 to September 2010. We match adults in the applicant households to their registered employment records, constructed from data provided by Uruguay's Social Security Administration (henceforth, SSA), which is responsible for collecting and recording payroll taxes and social security contributions from registered employment.We thus obtain a rich longitudinal database that covers all spells of registered employment for individuals in the program from January 2005 to December 2012. Because the records only reveal periods of formal (or registered) employment, we complement them with a detailed follow-up survey of eligible and ineligible households that applied to AFAM. The information collected in that survey enables us to observe (self-reported) informal work and non-employment for each household member.

Our identification of AFAM's effects relies on the program's eligibility poverty score, which is based on a household's predicted level of poverty as a function of its characteristics. The authorities strongly enforced this eligibility rule, creating a sharp discontinuity in the likelihood of participation at the cutoff point. The available evidence rules out manipulation of the assignment rule by applicants. Despite that discontinuity, we find some degree of imbalance in registered employment at baseline. We thus identify the impacts of the program through a difference-in-discontinuity research design that compares labor market outcomes for adults in applicant households just above (i.e., the treatment group) and just below (i.e., the comparison group) the program's poverty score eligibility threshold, controlling for potentially confounding pre-application differences around the threshold (Grembi et al., 2016; Bertrand et al., 2019).

We find that AFAM reduces formal labor force participation by about 6 percentage points (a 13% drop) among all beneficiaries, and by 8.7 percentage points (a 19% drop) for single mothers, who make up about 43% of the total. This translates into an elasticity of formal sector participation with respect to the net-of-tax rate of about 0.78 for the full sample and about 1.3 for single mothers. This effect persists for the two-year period following application to the program. We also break down the behavioral responses to the program by subgroup and by type of employment. We study heterogeneous impacts for individuals with different predicted levels of attachment to the formal sector in the absence of the CCT program and find substantial variation in this dimension. The reduction is stronger (8.9 percentage points, or a 29% drop) among individuals who have a medium propensity to be formally employed, with an elasticity of about 1.64, and is even stronger for single mothers in this group—a drop of 12.4 percentage points (about 50%), yielding an elasticity of about 3.46. The impact on those who have a low propensity to be formally employed is −2.5 percentage points and is not statistically significant, probably because these individuals would not be employed formally irrespective of their participation in the assistance program. For those who have a high propensity to engage in formal work, on the other hand, the effect of about −4.7 percentage points is substantially lower than the average effect in proportional terms (about 6%), with a low elasticity of about 0.31. The latter group's higher propensity to be formally employed implies that its formal labor force participation probably also would not be substantially affected by participation in the assistance program. We then present suggestive evidence concerning AFAM's impacts on informal employment and non-employment using data from a follow-up survey that sampled households on both sides of the discontinuity. We find that the decline in formal employment is divided more or less equally between individuals moving into inactivity and those moving into informal employment. Finally, based on our estimated behavioral responses, we provide efficiency cost calculations by computing the marginal value of public funds (MVPF) for AFAM single mothers. Despite pervasive informality in the context of the program, its MVPF of 0.61 implies an efficiency cost similar to that of cash transfer programs targeted to families in the United States, for which Hendren and Sprung-Keyser (2020) estimate an average of 0.74 and a confidence interval of [0.36,1.47], though lower than the MVPF of similar programs that induce labor supply disincentives, such as AFDC (MVPF of 0.87).

The main contribution of our paper is to provide clear and credible evidence of the negative effect of a CCT program on formal employment in a context of high informality. The existing literature reports small or null effects on the overall labor supply (Alzua et al., 2012; Banerjee et al., 2015), but the evidence on formal employment remains limited.[4] Garganta and Gasparini (2015) find that a large CCT program in Argentina reduced transitions to formal employment by 8.4 percentage points –a substantial drop of 40%. Their study reports intention-to-treat estimates, comparing poor households with and without children, and their empirical

[2] Gasparini and Tornarolli (2009) estimate that approximately 56% of wage earners in Latin America are informal workers—i.e., there are no payroll taxes nor social security contributions associated with their jobs.These workers are effectively excluded from social insurance benefits such as health and old-age pension coverage.

[3] We thank an anonymous referee for suggesting this organization of the discussion and some of the wording of the objectives.

[4] Besides the three papers cited in this paragraph, there are other studies and policy reports that provide suggestive evidence on CCTs and informality. In most cases, though, there are limitations in the underlying data or in the research design (see, for instance, Bosch and Manacorda, 2012; Araujo et al., 2017). In contrast, the empirical research on developed countries includes a sizable body of rigorous studies of the impact of social assistance on labor supply, as well as on the underlying mechanisms and magnitude of those impacts, the population groups most affected, and welfare implications (Chan and Moffitt, 2018). Moreover, these studies have generated a series of detailed analyses of social assistance programs'incentive effects on labor supply, which are often key to the design of welfare reforms aimed at improving the efficiency-equity trade-off (Blank, 2000; Scholz and Levin, 2001).

work relies on household survey data. Amarante et al. (2011), in turn, perform an exercise similar to ours for the PANES program in Uruguay (PANES was a more targeted, temporary, and smaller-scale predecessor to AFAM with eligibility also determined by a poverty score). The authors analyze the effect of the program on formal labor participation and earnings and find a negative impact on formal employment for men (but not for women) in eligible households of about 2.5 percentage points and of about 8 USD in monthly earnings. Bergolo and Galvan (2018), based on a sub-sample of the follow-up survey described below, study intra-household responses to AFAM and find a negative impact on the registered employment of married women. Bosch and Schady (2019) do not find effects of a CCT program on labor supply in Ecuador, although they report a small reduction in registered employment for women. These papers report some heterogeneous effects based on household and individual socioeconomic characteristics, but not based on an individual's propensity to be formally employed, which, as we discuss below, is a key aspect for the analysis and design of cash transfer programs. Moreover, the three papers report absolute and relative impacts of the programs on beneficiaries' labor supply, but fall short of providing efficiency cost estimates of the policies.

A second contribution is our analysis of the heterogeneity in responses. As stressed by Eissa et al. (2008), efficiency implications vary substantially when the analysis of impacts is broken down by subgroup. The literature on labor supply and on informality fails to properly acknowledge the heterogeneity in labor market attachment across workers in developing countries –in particular, the fact that some workers may not be at the margin of (in) formality –and the attendant implications for the impact of social programs. Gerard and Gonzaga (forthcoming) address this issue by exploiting heterogeneity across labor markets in Brazil and highlight the lower efficiency costs of unemployment insurance in labor markets with higher informality. Our analysis complements their findings with results from individuals within the same labor market.

A third contribution is the provision of efficiency cost impacts for the program. Existing papers on CCTs typically estimate impacts but fail to derive the implications of their results for welfare/efficiency. This point has also been recently emphasized by Gerard and Gonzaga (forthcoming), who develop a full welfare analysis of unemployment insurance in the context of informality in Brazil. Unemployment insurance programs are still small in Latin America, especially in comparison to CCTs such as AFAM. We compute the marginal value of public funds in what is, to our knowledge, the first application of Hendren's (2016) methodology to a developing country. This allows us to compare the efficiency implications of the AFAM program to those of cash transfer programs in the United States as reported by Hendren and Sprung-Keyser (2020).

In a broader context, this paper also contributes to a growing body of literature on public finance and development, particularly to studies that use administrative microdata and quasi-experimental methods to evaluate the efficiency cost implications of social policies in developing countries.[5] Our results add to this literature by extending its analysis to CCT programs, and by presenting efficiency cost implications.

The paper proceeds as follows. Section 2 describes the context of Uruguay's social protection system, the AFAM program, its rules and characteristics, and the expected effects of AFAM on participants' labor market outcomes. Section 3 describes the data sources and the construction of the datasets and samples we use in our empirical analysis. Section 4 discusses the empirical approaches on which we base our identification strategy. Section 5 presents the main results on registered employment responses to the program, their heterogeneity, the implied elasticities of participation in registered employment and the decomposition of the effects between informality and non-employment. Finally, Section 6 discusses the efficiency implications of the program's financial incentives. Conclusions follow.

2. The AFAM program: financial incentives and expected behavioral responses

2.1. Context of the AFAM program

Uruguay is a middle-income country in South America with a total population of approximately 3.3 million. The country has one of the oldest and most developed social protection systems in Latin America.[6] Access to most welfare and social insurance programs is linked to registered employment and financed through payroll taxes and contributions from both employers and employees, totaling about 32 percent of taxable wages. Registered (or formal) employees are those working in firms that reported them to the Social Security Administration (henceforth, SSA) and for which they paid the relevant taxes and contributions, which grants eligibility for social insurance benefits, mainly health insurance, family allowances, and old age pensions. A substantial fraction of employees are not registered with the SSA and thus are not covered by social insurance benefits. This phenomenon is referred to as labor informality (Gasparini and Tornarolli, 2009).

Uruguay suffered a severe economic crisis in 2002–2003. Unregistered workers, who lacked access to the counter-cyclical social assistance mechanisms provided by the SSA, were especially hard hit by this crisis. The government responded by launching a temporary conditional cash transfer program called *Plan de Atención Nacional a la Emergencia Social (PANES)*in 2005 targeted at the poorest 10 percent of households (see Manacorda et al., 2011 for details).

This emergency program was replaced in January 2008 by a new system of family allowances (Law 18.227), the *AFAM* program, as part of a broader progressive tax and transfer system reform. AFAM targeted poor households with children or pregnant women in the bottom 20 percent of the income distribution. It was implemented as a means-tested conditional cash transfer (CCT) program targeted to households in precarious socioeconomic conditions. The program's monetary transfers were conditionalon health checks (both for pregnant women and children) and school attendance for children in beneficiary households. AFAM became the most important social assistance program in Uruguay in terms of both the extent of coverage (about 42 percent of all children under the age of 18 in Uruguay) and the magnitude of the cash benefits provided (about 0.35 percent of GDP).

2.2. Eligibility and disqualification from AFAM

Eligibility for AFAM depended on two criteria: the household had to pass both an income test and a proxy means test. To participate in the program, households were first required to complete an application, in which they provided an array of socioeconomic information, including household characteristics (address, housing conditions, dwelling type, characteristics and quality, ownership, access to water and sanitation, etc.), and detailed information about household members, such as their national identification number, education levels, labor force participation, and income levels.

[5] See for instance Kleven and Waseem (2013), Best et al. (2015), Pomeranz (2015), Carrillo et al. (2017), Gerard and Gonzaga (forthcoming), Naritomi (2019).

[6] See Appendix A.2.1 for more details on Uruguay's social insurance system and the context of the AFAM program.

After completing the application, households were first subject to an income test: the per capita income of the household had to be below a predetermined threshold in order to qualify for the program. The household's income level for this test was computed by combining the information reported on the application form and SSA administrative records. The SSA computed total household income by matching its administrative earnings records with each household member's national identification number. These earnings included those from registered employment (as reported to the tax and social insurance authorities by employers), from pensions (paid by the SSA), and from other government transfers recorded in the SSA administrative records. According to the information provided by AFAM authorities, about 70% of the total verifiable household income corresponded to earnings from registered employment. We refer to this as "verifiable income" throughout the rest of the paper. The income test compared per capita household income with a given threshold. Crucially, unlike programs in other countries in the region (such as the Asignación Universal por Hijo in Argentina –Garganta and Gasparini, 2015), AFAM was compatible with formal employment (up to a certain level of earnings determined by the income test threshold). That is, formally employed individuals could still be eligible to receive AFAM cash transfers provided their income from formal employment was sufficiently low.

The second step for determining eligibility after the income test was a proxy means test. This test relied on a poverty score calculated by program officials and was based on a large set of socioeconomic characteristics provided by the household on the application form. The score's algorithm was devised in consultation with academics and social policy experts and details of its components and construction were never disclosed to the public.[7] Households with income below the income test's threshold and with a poverty score above a predetermined level were deemed eligible to participate in the AFAM program. Households could still be rejected or disqualified for administrative reasons, such as, for example, if they failed to submit proof of children's school enrollment or of completion of health checks. Appendix Table A.4.1 summarizes reasons for rejection of AFAM applicants. About 50 percent of the applicant households were deemed ineligible because they failed the income test, the proxy means test, or both.

Most eligible households ended up participating in the program. The program rules specified three main situations that could lead to a household's disqualification after enrollment. First, a household's earnings computed from administrative records (its verifiable income) should not rise above the income eligibility threshold. The income test was applied to participant households every two months, and it was strictly enforced. Second, households had to adhere to the mandatory health check and school attendance requirements for children. Finally, households were disqualified from the AFAM program when all of the household's children reached 18 years of age. Our analysis of the 2009–2012 administrative records of AFAM's disqualifications, presented in Appendix Table A.4.2, indicates that, on average, about 15 percent of AFAM beneficiary households were disqualified each year during that period, and about 57 percent of disqualifications were attributable to households' verifiable incomes exceeding the income test threshold.

2.3. AFAM monetary benefits and implicit financial incentives

Eligible households that participated in the program were entitled to a monthly cash transfer. A key feature of the AFAM program's incentives was that 100 percent of verifiable income up to the threshold amount for program eligibility was exempt from taxation, which yields an implicit marginal tax rate on the benefit of zero percent. Above this threshold, benefits abruptly drop to zero, creating a "notch" in households' budget constraints.[8]

The amount of the transfer varied according to the household's number of children under the age of 18 and the number of children in the household attending secondary school. The transfer was larger for those in secondary education to encourage older children to attend and finish school.

The total benefit granted to a household was computed as follows:

$$G_{AFAM} = \begin{cases} 0 & \text{if } (1-\theta)\times Y^F > T \\ \beta \times (Kids0to17)^{0.6} & \\ +\delta \times (HighSchoolKids)^{0.6} & \text{if } (1-\theta)\times Y^F \\ \leqslant T \text{ where } \theta = 0.15 & \end{cases} \quad (1)$$

where Y^F is per capita household gross formal (or verifiable) income, T is the per capita income test threshold, θ is a fixed deduction equal to 15 percent of the household's gross verifiable income, $Kids0to17$ represents the number of children below 18 years old, $HighSchoolKids$ the number of children that attend secondary school, and β and δ are transfer levels.

The cash transfer and the income test's threshold levels were adjusted periodically according to the evolution of the official Consumer Price Index. The resulting cash transfer levels were relatively generous sufficiently so to affect beneficiary households' labor supply decisions: for 2008–2012, the transfer amounted on average to about 68 percent of the minimum wage for an average AFAM household and about 48 percent of the poverty line (Appendix A.2.3 details the levels of the parameters).

All households in our study sample passed the income test at the time of application and were deemed eligible or ineligible according to the proxy means test. The latter was only conducted at the time of the original application and was not subject to manipulation. On the other hand, beneficiary households could adjust their labor supply and were subject to the income test every two months. Our discussion focuses on this frequently enforced eligibility condition. The level of the benefit and the income test constitute the two key factors for understanding labor market responses to AFAM in the context of our paper.

Fig. 1 illustrates the program's implicit financial incentives to engage in formal employment. It depicts the simulated budget set for a representative individual in our setting –a single mother of one, beneficiary of AFAM, whose child attends secondary school (single mothers represent about 43% of AFAM adult beneficiaries). The tax schedule for most low income registered workers in Uruguay is essentially flat and corresponds to a payroll tax amounting

[7] The eligibility poverty score was devised by researchers at the Universidad de la República (UDELAR) in Montevideo, Uruguay (see Amarante and Vigorito, 2011). The algorithm is based on the coefficients of a probit model in which the dependent variable is equal to one if a household is within the first—i.e., the lowest —quintile of per capita income, and zero for those above the first quintile but below the median per capita income. The original model was estimated by means of a fully saturated function of household variables drawn from the ECH national household survey. The resulting coefficient estimates were used to predict the score for each applicant household based on data from the application form, and the eligibility thresholds for residents of Montevideo, the country's capital, were different from those for the rest of the country, to reflect differences in living costs. Uruguay's Social Security Administration,*the Banco de Previsión Social* (BPS), and the Ministry of Social Development (*Ministerio de Desarrollo Social*, henceforth MIDES), were responsible for computing the score and monitoring household income and conditionalities.

[8] This feature of the AFAM program is similar to most of the income support programs in Latin American countries, though not to traditional welfare programs in developed countries where income benefits are reduced at a specific rate. The benefit structure of the Job First program in Connecticut was similar to AFAM's structure in that program rules exempted from taxation 100 percent of earnings up to the monthly federal poverty line (Bitler et al., 2006; Kline and Tartari, 2016; Hartley and Lamarche, 2018).

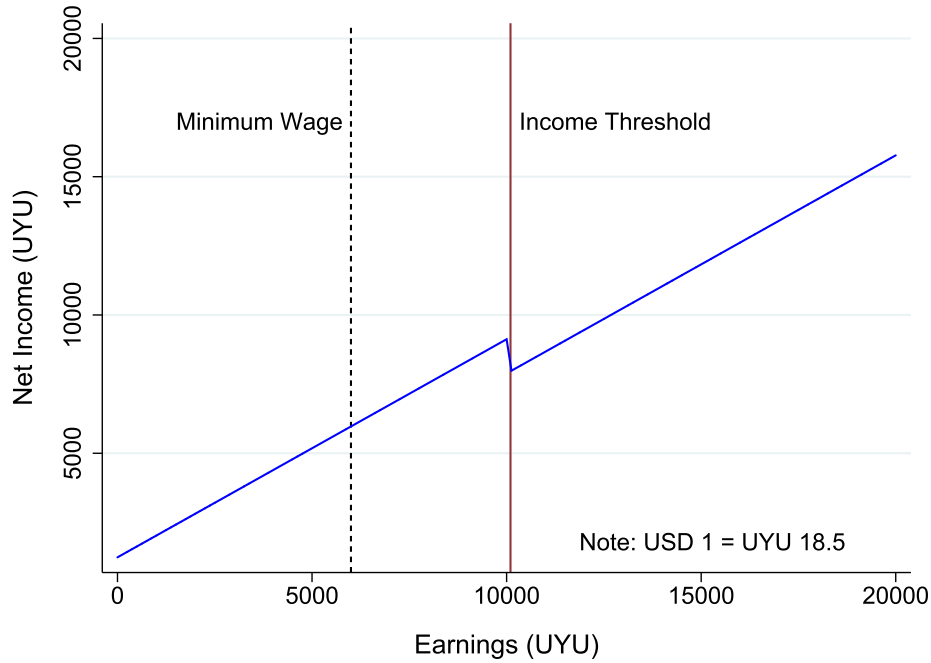

Fig. 1. Budget Set for Single Mother with one Child.
Notes: This figure presents a simulation of the stylized budget set for a single mother with one child. The dataset corresponds to the *AFAM administrative records* and Uruguay's ECH household survey for the year 2011. The figure is based on the AFAM cash benefit, AFAM income threshold, national minimum wage, individual earnings, and the nominal exchange rate for the year 2011. The AFAM cash benefit is computed using the following Eq. (1): $UYU1236 = 865 \times (1)^{0.6} + 371 \times (1)^{0.6}$. A household's income-eligibility threshold is transformed to an equivalent "per-person" amount to make it comparable with individual earnings –i.e., $T^* = T \times HHmembers/(1-\theta)$ = $(4287 \times 2/0.85) = UYU\,10,088$.

to 21 percent of earnings –21 percentage points out of the 32 percent total payroll tax is payed by the employee. The AFAM cash benefit is computed according to Eq. (1) with parameters corresponding to 2011. As a benchmark, the graph shows the level of the national minimum wage.

The AFAM income threshold for eligibility generates a large disincentive to work formally. There is a drop in net income as gross verifiable earnings exceed UYU 10,088 because at this earnings level the AFAM benefit is lost entirely.[9] If she engages in formal work, at an earnings level of UYU 10,088, the single mother in the example depicted in Fig. 1 loses UYU 2118 (10,088×0.21) of her income in net tax payment and UYU 1236 from the withdrawal of AFAM benefits. That is, the net income gain from working at a registered job declines from UYU 9206 to UYU 7970. The effective average tax rate relevant to the decision to engage in formal employment at this level of registered earnings is 33.2 percent –i.e., 21 percent from the tax and an additional implicit tax rate on registered employment of 12.2 percent (UYU 1236/10,088) as a consequence of losing the cash transfer.

2.4. Expected behavioral responses to AFAM and efficiency implications of the program

In a labor market with high informality, as is the case in most developing countries, the standard labor supply model predicts that an income-tested transfer program such as AFAM could induce a reduction in potential beneficiaries' registered employment at the extensive margin, resulting in either higher non-employment or higher informal employment.[10]

Our analysis of AFAM's effects on labor supply is based on a comparison of households that passed the income test at the time of application but were deemed either eligible or ineligible by the proxy means test (i.e., their eligibility poverty score). Enrolled households' post-application earnings from registered employment were monitored continuously and failure of the income test resulted in cessation of the benefit.[11]

We can illustrate the effects of the program with the cases of two comparable adults from similar households that applied to the program at the same time, passed the income test, but fell on different sides of the eligibility poverty score threshold. The adult in the household that passed the proxy means test and became a beneficiary of the program was more likely to limit her participation in registered employment because of an income effect from relatively higher unearned income. The adult in the beneficiary household also experienced a substitution effect that impelled her to limit her participation in registered employment because higher formal earnings could result in loss of the benefit due to the continuous formal earnings-based income test. The reduced formal labor participation in both cases could result in informal work or non-employment. The adult in the ineligible household, on the other hand, was more likely to enter or increase participa-

[9] While the income-eligibility threshold is computed in terms of household per-capita (formal) income, for the sake of simplicity we render it equivalent to a "per-person" rate to allow for comparison with individual earnings. Specifically, $T^* = T \times HHmembers/(1-\theta)$ = $(4287 \times 2/0.85) = UYU\,10,088$, where *HHmembers* corresponds to the ratio of a household's total number of members to the number of formally employed individuals. In this example, *HHmembers* is equal to 2 because we assume the mother is the only formally employed member in the household.

[10] In fact, AFAM's design implies that it might also affect the intensive margin of labor supply, because it might generate a notch at the income test threshold in households' budget constraints (Kleven, 2016). Unfortunately, we are unable to measure labor supply or misreporting at the intensive margin because our data do not cover working hours or earnings (see Section 3).

[11] Beyond the standard labor supply arguments, the program's conditionalities (requirements in terms of children's health checks and school attendance) also might have effects on the labor supply of adults in potential beneficiary households. We discuss these additional considerations in Appendix Section A.4.1.

tion in formal employment, especially in the context of a growing economy as in the period under study.[12]

Behavioral changes induced by AFAM might be due either to substitution or income effects and might result in increased non-employment or, in a setting with a high level of informal employment, in more informal work.The total share of eligible adults in registered employment should decrease. Moreover, as long as there are two tiers of informality in the labor market (Fields, 2009), the negative effect on registered employment should be higher among individuals with a higher propensity to be formally employed than on potential beneficiaries with a lower propensity to be formally employed (see the discussion of these potential heterogeneous effects in Section 5.2).

These effects can be used to evaluate the efficiency impact of the program. Standard public finance analysis indicates that substitution effects lead to distortions and thus have efficiency costs, whereas income effects do not generate distortions at the margin. Only the former matters for understanding policies'efficiency implications. Our reduced-form estimates of the program's effects on employment do not allow us to separate between the income and substitution effects at play in the behavioral responses to AFAM. This imposes a clear limit on the interpretation of our results in terms of traditional efficiency cost analysis. In Section 6, we present an alternative to assess the efficiency implications of the program following Hendren's (2016) "policy elasticity" approach. This framework relies on a policy's effect on government revenue as a sufficient statistic for all behavioral responses, and posits that the welfare analysis of public policy changes can be based on individuals'willingness to pay out of their own income. The only behavioral response required to compute the efficiency cost impact of the program is the causal effect of the AFAM on the government's budget, which does not require computing the compensated behavioral responses of traditional welfare analysis.[13] We can thus bypass the impossibility of distinguishing between income and substitution effects in our setting.

3. Data sources and sample construction

Our main analysis is based on a series of administrative and household survey datasets matched by means of a unique national individual identity number, the "Cédula de Identidad."Every citizen is assigned one of those numbers at birth, and it is required (and recorded) in all transactions related to registered employment, social insurance and welfare programs. This section provides a brief description of the original data sources, their characteristics and time frames. It also documents the matching process, the resulting datasets used in our empirical work, and the outcomes of interest. For more details see Appendix Section A.1.

3.1. Data sources

AFAM Administrative Records. Our primary dataset is the *AFAM administrative records*, which includes individual- and household-level data on application to the program and during participation in it. This dataset was constructed from two sources. The first, the *AFAM baseline application records*, consists of responses to a detailed questionnaire on the socioeconomic and demographic characteristics of all individuals in households that applied to the program. This dataset also includes the date of application and, most importantly, the exact value of the household's eligibility poverty score computed by the authorities –we standardize that score for our analysis.[14] This rich baseline data contains information for both successful and unsuccessful applicant households, and it covers the period that runs from January 2008 to September 2010. The second dataset, the *AFAM participation records*, corresponds to the participation history of the individual/household once they were enrolled in AFAM –i.e., it covers a sub-sample of households in the *AFAM baseline application records*. This dataset covers the period January 2008 to March 2012. This dataset was compiled by the program's implementation institutions, MIDES and the SSA.

SSA Registered Employment Records. Our main outcome of interest is registered (or formal) employment. We obtain information on this outcome from a further source, the *SSA registered employment records*, which keeps monthly records of monetary contributions made by employers and employees to compute eligibility for and levels of social insurance services and transfers.[15] A formal employee is one who is "registered" with the SSA, and thus is covered by the social insurance benefits provided by the SSA. We have access to these records for program applicants for the period that spans from January 2005 to December 2012. The main advantage of this type of data is that it records all episodes of registered employment for both employees and self-employed workers, in both the private and public sectors. We construct a longitudinal database of monthly registered employment histories that covers the entire period under study and, most importantly, substantial periods of time before and after the application records. In terms of limitations, the *SSA registered employment records*do not include information on hours worked per day (or days per month) nor on earnings from registered work, which restricts the scope of our analysis. We cannot, for instance, determine the impact of AFAM on the intensive margin of labor supply or fully estimate all elasticities (see the discussion in Section 5.3).

AFAM Follow-Up Survey. An additional limitation of administrative databases for the study of labor market outcomes in developing countries is that, by definition, these sources do not have any information on unregistered (or informal) employment. Therefore, to complement the administrative data source, we use a follow-up household survey (the *AFAM follow-up survey*) specifically designed to study the effects of AFAM on household welfare and on individual labor market responses.[16] The survey's sampling frame was designed taking into account the discontinuity in the program's eligibility score threshold with the purpose of exploiting the quasi-random variation generated by the eligibility rule. It was based on a stratified random sample of eligible and ineligible households around the cutoff point.The interval of the (standardized) eligibility poverty score for the survey was set in the range $[-0.0426; 0.0727]$. The *AFAM follow-up survey* was conducted from September 2011 to February 2013. Its questionnaire consisted of a shortened version ofUruguay's National Household Survey (the *Encuesta Continua de Hogares,* henceforth ECH). It asked extensive questions about household living conditions, socioeconomic charac-

[12] Uruguay's economy grew after the introduction of AFAM, with an increase in the overall rate of formal employment in the period under study(see Fig. A.2.1 in Appendix A.2.2), which implied better labor market prospects for potential beneficiaries. The impact we estimate in our empirical work below thus corresponds to a relatively smaller increase (rather than an absolute decrease) in formal employment.

[13] The envelope theorem implies that behavioral responses to marginal policy changes do not have a direct impact on individual utility, and thus the welfare analysis can rely solely on a policy's causal effect on government revenue (Hendren, 2016).

[14] We standardize this score so that the eligibility threshold is zero.The standardized eligibility score for the population of all applicant households is in the range $(-0.257; +0.712)$. Eligible households have positive values and ineligible households have negative values.

[15] As detailed in Section 2, the SSA provides old age pensions, unemployment benefits, and maternity and child allowances, among other transfers, and determines health insurance eligibility.

[16] The survey was designed and implemented by researchers at the Instituto de Economía (IECON) of theUniversidad de la República (UDELAR), in collaboration with MIDES and researchers at the Institute of Statistics and at the Department of Sociology at UDELAR (Amarante and Vigorito, 2011).

teristics and, crucially, mutually exclusive labor market outcomes on the date of the interview –i.e., registered and unregistered employment, unemployment and non-participation –for each adult individual in the sample.

Encuesta Continua de Hogares-ECH. We also use the microdata from the ECH for the period 2008–2012. The ECH is a nationally representative household survey conducted according to international standards. It is carried out periodically by Uruguay's national statistical agency (*Instituto Nacional de Estadísticas*, INE). We use ECH data to describe labor market patterns in Uruguay (see Appendix Section A.2.2) and to impute income for estimating the participation tax rates (see Section 5.3).

3.2. Samples for the empirical analysis

From these multiple data sources, we construct two related but distinct samples and datasets for our empirical analysis.

Main Sample. We first matched the *AFAM administrative records* with *SSA registered employment records* to yield complete registered work trajectories for the period that runs from January 2005 to December 2012 for all adults in households that applied to AFAM between January 2008 and September 2010. The matched dataset also contains AFAM participation history for the same period and sample. We adjust this data to reflect the fact that application to AFAM occurred at different moments during the period under study (Appendix Section A.2.4 discusses patterns of AFAM application and participation over time). As in an event study, we re-center the work histories at the time of application to the program for individuals in both eligible and ineligible households.

We impose four restrictions to construct our study sample. First, we include only household heads and their spouses (when present) who were aged 18 to 57 at the time of application to AFAM during the period January 2008 to September 2010. These age limits restrict our sample to the economically active population over the whole period under study.[17] Second, we restrict the sample to the population of applicants in households that fall in the eligibility poverty score range $(-0.257; +0.257)$.[18] Third, we restrict our observation window for the data on registered employment to 36 months before and 24 months after the household's application date in order to generate and analyze a balanced panel. While we could study longer post-application periods for some individuals, our criterion balances the length of the pre- and post-application periods for all individuals in our sample, and thus avoids the selection issues that would arise if we mixed in our sample individuals from households with different lengths of enrollment. Fourth, to use a homogeneous sample across all of our empirical work, we discard a small number of observations (fewer than 5 percent) for which we cannot impute labor income due to missing covariates since this imputation is required to compute the elasticity of registered employment (see Section 5.3).[19]

From the population of 240,146 adults in all applicant households in the range $(-0.257; +0.712)$ of the eligibility poverty score, we use a sample (henceforth, the *Main Sample*) of 83,591 individuals (74,652 eligible and 8939 ineligible) in households in the eligibility poverty score range $(-0.257; +0.257)$. For some specific analyses, we report results based on the $(-0.257; +0.712)$ bandwidth of the eligibility score (we refer to this as the *Population*). With these restrictions, the *Main Sample* consists of a five-year balanced panel of 5,099,051 monthly observations from 83,591 individuals observed for 61 months each (36 months pre-application, the application month, and 24 months post-application).

Follow-Up Sample. The second sample (henceforth, the *Follow-Up Sample*) consists of adults in households interviewed for the *AFAM follow-up survey*during the period September 2011 to February 2013. As described above, these households were also drawn from a range around the program's poverty eligibility score. There were 2403 adults (18–57) interviewed in the follow-up survey (1656 eligible individuals and 747 ineligibles). We matched the survey responses from these individuals with their and their households' information (supplied by the*AFAM administrative records* and the *SSA registered employment records*. Table A.1.1 in the Appendix presents summary statistics of selected socioeconomic characteristics for the samples we use in the empirical analysis.

3.3. Main outcomes of interest

The main outcome of interest in our analysis is registered employment as recorded in *the SSA registered employment records*. Specifically, registered employment in our data is an indicator variable equal to '1' if the individual is registered with the SSA in a given calendar month, and zero otherwise.

The analysis in Section 5.4 is based on the *AFAM follow-up survey*, and we exploit the information from the survey to explore responses to AFAM on additional margins of labor force participation. The first of those margins is registered employment, based on responses to a specific question in the follow-up survey and defined as an indicator coded as '1' for individuals who declared themselves to be a registered employee at the time of the interview and zero otherwise.[20] The second outcome of interest in our study is unregistered employment, again as reported by the respondent in the follow-up survey. This indicator is coded as '1' for individuals who work but state that they are not registered with the SSA, and zero otherwise. The third outcome is non-employment, which is an indicator variable equal to '1' when the individual declares that she is not working, and zero when she states that she is employed (either as a registered or as an unregistered worker).

The variable that defines assignment to the treatment in our empirical analysis is the eligibility condition for enrollment in AFAM. As discussed below, for identification reasons eligibility for AFAM is defined according to the household's eligibility poverty score (henceforth, "eligibility score") during the process oftargeting household applicants. We standardize this score so that the eligibility threshold is zero, with positive values for eligible households and negative values for ineligible households.

4. Empirical approach

Our empirical strategy exploits a specific feature of the AFAM program's eligibility rules. Households that apply to the program must first pass an income test based on household members' earnings from registered employment, as reported to tax and social security authorities. As described above, when household income is below a certain threshold (i.e., when the household passes the income-eligibility test), the household is subject to a proxy means test and assigned a eligibility score based on a detailed set of

[17] The lower age limit means that individuals under age 18 would not have benefited from the program as children (e.g., they were not required to be enrolled in school). The upper age limit of 57 years reflects the fact that workers are eligible for retirement benefits in Uruguay once they reach 60.

[18] Restricting the sample to households symmetrically located around the zero value does not alterthe choice of the optimal bandwidth (Calonico et al., 2014), but it does substantially reduce the computational time for obtaining our estimates.

[19] Our main results concerning the program's impact on registered employment remain qualitatively the same and quantitatively very similar if we include these discarded observations (comparison of results in Table 1 and Appendix Table A.6.8).

[20] The specific question in the ECH and in the *AFAM follow-up survey* is: "Are you contributing to a retirement benefit through your current job?" (in Spanish, "Aporta a alguna caja de jubilaciones por su trabajo actual?"). This is a standard question to gauge registered or formal work in household surveys in Latin America (see, for instance, Gasparini and Tornarolli, 2009; and Galiani and Weinschelbaum, 2012).

socioeconomic and demographic characteristics. The household is deemed eligible for the program only if its score is above a predetermined threshold. The authorities strongly enforced this eligibility rule, which created a sharp discontinuity in the likelihood of participation at the cutoff point as depicted in Fig. 2. The figure plots the proportion of applicant households (vertical axis), both those deemed eligible and those deemed ineligible through the application process, that were enrolled in the program at any given point in time since its implementation in 2008 until the end of our observation window in 2012, as a function of the eligibility score (horizontal axis). The figure clearly shows a sharp discontinuity in the probability of participation in AFAM at the threshold. Participation is about 96 percentage points higher for households with a score just above the eligibility threshold compared to those with a score just below it.[21] Thisfinding confirms that the AFAM eligibility score assignment rule was strictly enforced.

Fig. 3 depicts the level of our main outcome of interest (registered employment) around the threshold before and after program application. In each panel in the figure, we plot for the *Main Sample*the mean unrestricted monthly registered employment (blue circle) in bins of width equal to 1 percentage point of the score and the estimated monthly means (solid red line) from alocal linear regression applied to each side of the cutoff point along with the 95-percent confidence interval. Panel Ain Fig. 3 depictsthe pre-application difference in registered employment around the AFAM eligibility threshold. There is a small but statistically significant jump at the threshold of about −3.8 percentage points (the RD estimate, discussed below, is reported in Table 1). Registered employment levels for the two groups start to diverge substantially after application to the AFAM. Panel B in Fig. 3 shows a much larger jump at the eligibility threshold of about −9.4 percentage points (also reported in Table 1).[22]

Despite the sharp discontinuity in eligibility depicted in Fig. 2, the imbalance in the pre-application period main outcome, and in some other baseline characteristics (as documented and discussed in Appendix Section A.4.3) may signal manipulation of the running variable, which would compromise identification of causal effects in the context of an RD research design based on Fig. 3. If individuals were able to manipulate the program's eligibility process, the difference between the eligible and the ineligible at the threshold would reflect some systematic advantage favoring the eligible, instead of being determined by plausibly exogenous idiosyncratic factors. This would happen, for instance, if program officials favored households with adults engaged in informal jobs, or if applicants lied about their socioeconomic characteristics when filling in the application form as a strategy to gain eligibility. Either situation would introduce systematic differences between observable and unobservable characteristics of individuals just above and just below the eligibility threshold. Appendix Section A.4.3 discusses the results of a McCrarytest –i.e., the density of the eligibility score and a smoothed density estimator based on a local linear regression on both sides of the threshold and other auxiliary evidence –which indicate that despite the imbalance in some characteristics we cannot reject the null hypothesis of no discontinuity. Finally, besides this empirical evidence, the institutional context of AFAM also suggests the absence of selective sorting or of eligibility score manipulation by either beneficiaries or program officials. While individuals had incentives to complete the application form strategically to gain eligibility, they were limited by the fact that the government did not disclose the algorithm used to compute the score, the characteristics on which it was based, their underlying weights, or the level of the eligibility threshold. Moreover, the very large, sharp jump in participation at the threshold illustrated in Fig. 2 indicates that the program's rules were strictly enforced.

While there are alternative ways to compute bounds in cases of a manipulated running variable in the context of RD (Gerard et al., 2020), the empirical evidence does not point towards a case of manipulation. However, the imbalance in the main outcome (and in some of the covariates) may confound the impact of the program with these pre-application differences. We thus use an alternative to the standard RD model that exploits the longitudinal nature of our data in an RD framework to control for pre-application differences. Specifically, to identify the impacts of the program, we use a difference-in-discontinuity (DD-RD) research design that compares the discontinuity in post-application outcomes to the discontinuity in pre-application outcomes for individuals close to the threshold, thus controlling for potentially confounding pre-application differences around the threshold (see e.g., Grembi et al., 2016; Bertrand et al., 2019).[23] The DD-RD specification is similar to the standard RD estimation, but it also includes information from the pre-application period. Specifically, the DD-RD model that we estimate is given by the following equation:

$$Y_{it} = \alpha_0 + \beta_0 ELEG_i \times Post_t + \beta_1 ELEG_i + \beta_2 Post_t + \tag{2}$$

$$f(score_i)[\kappa_0 + \kappa_1 Post_t + \kappa_2 ELEG_i + \kappa_3 ELEG_i \times Post_t] + \lambda_t + \epsilon_{it}$$

where Y_{it} is the outcome of interest for individual i at time t; $ELEG_i$ is a dummy variable equal to 1 if the individual belongs to an applicant household eligible for the program (i.e., if $score_i$>0), and zero otherwise;*score*is the value of the eligibility score, which in the RD literature is standardized relative to the eligibility threshold (c); $f(score_i)$ is an (unknown) functional form of the "assignment" or running variable*score*; and λ_t represents time fixed effects. Each variable's interactions with a post-application (*Post*) indicator represent the time dimension that the DD-RD model adds to the standard RD analysis.[24] The parameter β_0 captures the causal effect of AFAM on the outcome of interest. In the graphical interpretation, this coefficient is the difference between the jumps in registered employment at the discontinuities in panels A and B in Fig. 3.[25]

The functional form of $f(.)$ and the window on each side of the cutoff threshold are key inputs for RD-based research designs.[26] Our baseline specification estimates this function using local linear regressions with a triangular kernel density for a given bandwidth (Cheng et al., 1997; Porter, 2003). When the analysis focuses on the*Main Sample*, we implement the bandwidth selection process following Calonico et al. (2014). Estimates based on the *Follow-Up Sample*use all of the survey's observations –the sample was drawn from an RD optimal bandwidth.We report MSE-optimal coefficients

[21] Only a handful of households with a score below the cutoff participated in AFAM.

[22] It should be noted that registered employment fluctuates in the 30–40 percent range in the pre-application period, and in the 40–60 percent range in the post-application period. This is due to the fact that the program was launched when Uruguay's economy was recovering from the severe crisis of 2005, after which formal employment rose substantially. The pre-application period is closer in time to the crisis than the post-application period, and the difference in average registered employment for applicants in panels A and B in Fig. 3 reflects the employment recovery during the period we study. Macroeconomic and labor market trends during our period of analysis in Uruguay are discussed in Appendix Section A.2.2.

[23] Additional earlier studies based on RD designs that exploit longitudinal data include Lemieux and Milligan (2008, 2010, 2012), among others.

[24] Some subsets of our results are based on conventional RD estimates (for instance, those based on the follow-up survey, for which we only have one observation in time) which correspond to Eq. A.4.1 –i.e., the Eq. (2) excluding the $Post_t$ terms and the corresponding interactions.

[25] We implement the DD-RD estimator as follows: We first estimate Eq. A.4.1 using data from the pre-application period (36 months); we then subtract the RD regression coefficient from the outcome variable for those eligible, and we finally estimate the RD model using the post-application period (24 months). The RD coefficient of that regression corresponds to β_0 in Eq. (2).

[26] See Lee and Lemieux (2010) for a detailed review of technical issues relevant to the RD research design.

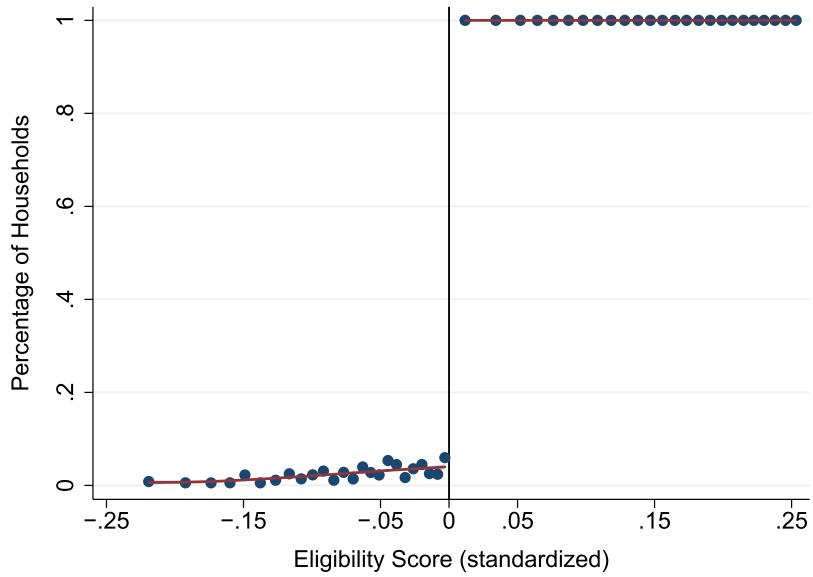

Fig. 2. The AFAM Intake Process: Participation Rates and Eligibility Score.
Notes: This figure plots participation in AFAM against the eligibility score for the sample of households with heads, and spouses of heads, aged 18 to 57 at the time they applied to the program. The dataset corresponds to the *AFAM administrative records* (January 2008-September 2010). The eligibility score is standardized so that the eligibility threshold is zero, with positive scores indicating individuals in eligible households and negative scores indicating those in ineligible households. Each point soos (solid circle) in the plot represents the percentage of individuals in applicant households who have participated in AFAM in eligibility score bins of one percentage point. The solid line plots estimated means from a local linear regression applied to each side of the eligibility threshold.

and standard errors (referred to as "conventional estimates" by Calonico et al., 2014), but rely on bias-corrected estimates for p-values, since these are optimal for inference in this context.[27] For estimates from the longitudinal *Main Sample,*we report robust standard errors clustered at the household level to account for potential serial correlation of outcomes for adults over time.[28] When we use the *Follow-Up Sample*, we report conventional robust standard errors.

The validity of our DD-RD design rests on the assumption that unobserved characteristics of eligible and ineligible individuals evolve similarly across time at the eligibility cutoff point. An indirect test of this assumption would be to check whether the discontinuities in observed characteristics, in the eligibility score, or in the outcome of interest change over time. Unfortunately, the covariates and the running variable are time-invariant in our setting –i.e., we observe this information only at the baseline.[29] However, given the longitudinal nature of our *Main Sample*, we can test whether there are differing discontinuity patterns around the eligibility threshold in our outcome of interest throughout the pre-application period. Reassuringly, inspection of Fig. 4 (discussed in detail in the following section) indicates that the difference in registered employment at the threshold between eligible and ineligible individuals is roughly constant over the pre-application period.

5. Registered employment response to the program

5.1. Registered employment response to AFAM

The essence of our main result is illustrated non-parametrically in panels A and B in Fig. 3. AFAM generated a substantial decline in registered employment among its beneficiaries. In this section, we analyze this effect in greater depth. We first turn,in Table 1, to the detailed statistical results of the average effects of AFAM on registered employment over the whole period using the *Main Sample* of analysis. Column (1) reports the point estimates, while column (2) presents the optimal bandwidth used in the estimation following Calonico et al. (2014). Column (3), in turn, reports the average registered employment rate for ineligible individuals close to the threshold for the left-hand side of the bandwidth (which we label as our comparison mean). Column (4) presents the effects in column (1) as a proportion of the registered employment rates in column (3), while column (5) reports the number of observations (individuals by time). Rows (1) and (2) report estimates based on the RD specification given by Eq. A.4.1 using pre- and post-application data respectively. In turn, row (3) reports the main estimation results based on the DD-RD specification given in Eq. (2). All the estimates include time and date-of-application fixed effects to control for different potential underlying trends between eligible and ineligible groups. Panel A presents the estimates without socioeconomic controls, while panel B presents equivalent results, adding a standard set of covariates to the basic econometric models.[30]

[27] The authors thank Gonzalo Vázquez Baré for this suggestion.

[28] We discuss program eligibility and income conditionalities at the individual level, although program participation and conditionalities were actually determined at the household level. We cluster standard errors at the household level in our empirical analysis to make up for this. However, the incentives to work or to not work, and to work formally or informally, may be a function of the combined responses and earnings potential of all household members (Galiani and Weinschelbaum, 2012). The elasticities we estimate in Section 5.3 might be biased if earnings potential is correlated across household members, although only about 10 percent of adults in our study sample belong to the same household.

[29] The results presented in the following sections are computed and presented with and without covariates to control for these baseline differences in characteristics. The results do not change significantly with the inclusion of these covariates.

[30] The controls in the regressions are those described in Table A.1.1. They include individual characteristics such as gender, head of household status, age, marital status, educational level (in three categories), the number of children in the household aged 0–17, whether the household was enrolled in the PANES program, residence in Montevideo (Uruguay's capital), and whether the individual was employed. These covariates allow us to control for the imbalances documented in Table A.4.3, and they also allow us to gain precision by reducing the residual variance.

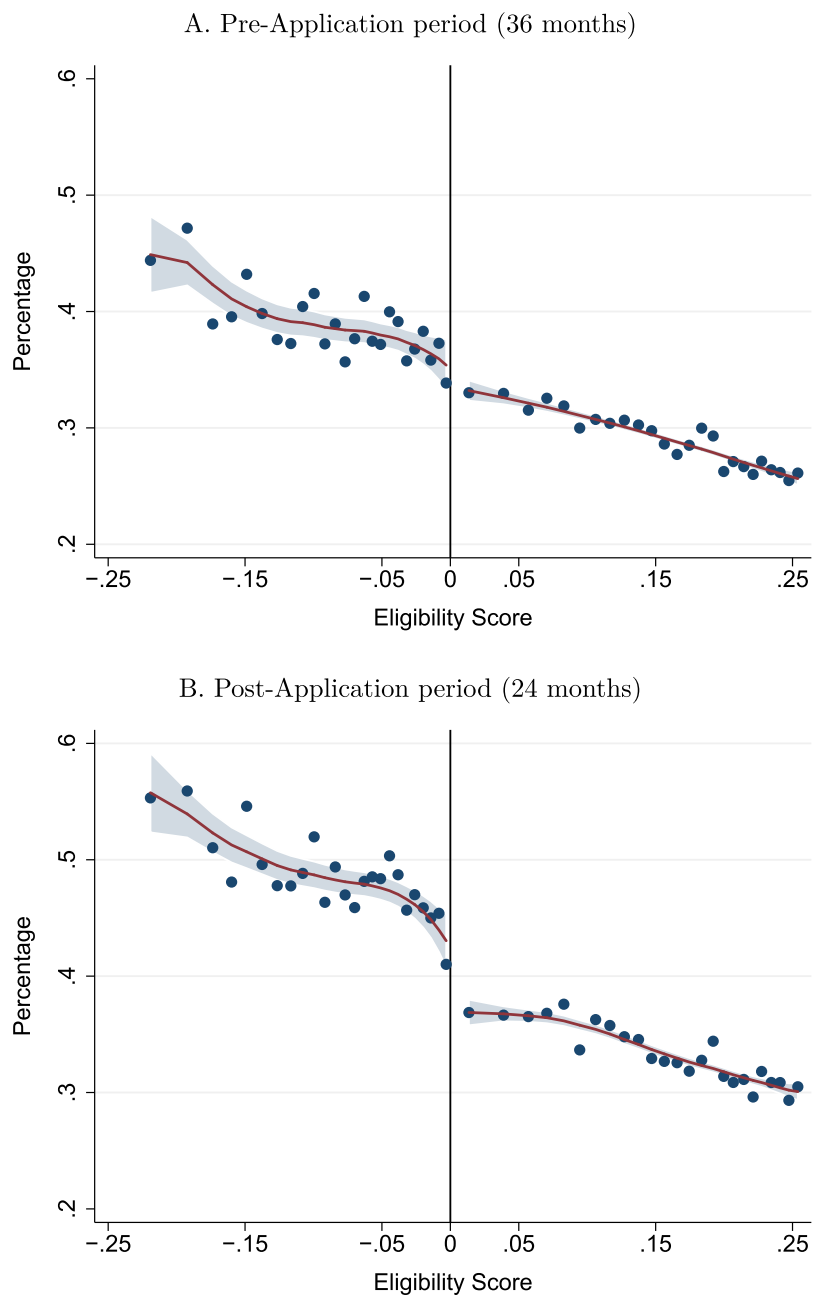

Fig. 3. Difference-in-Discontinuity: Registered Employment Rates by Eligibility Score, Pre- and Post-Application.*Notes*: This figure plots registered employment against the eligibility score. The sample corresponds to heads of households, and spouses of heads of households, aged 18 to 57 at the time of application to AFAM during the period January 2008-September 2010. The dataset corresponds to the *AFAM administrative records*matched with the *SSA registered employment records* for the pre-application (36 months) and post-application (24 months) period according to the SSA's administrative records (see Section 3.2 for a detailed description of the data). The eligibility score is standardized so that the eligibility threshold is zero, with positive scores indicating individuals in eligible households and negative scores indicating individuals in ineligible households. Each point (solid circle) in the plot represents the average value of the outcome variable in eligibility score bins of one percentage point. The solid line plots estimated means from a local linear regression estimated at each side of the eligibility threshold, along with the 95-percent confidence interval (shaded area).

The RD estimates in panel A show a large and statistically significant decline of 9.4 percentage points in the registered employment of adults eligible for AFAM relative to the rate for ineligible adults in the post-application period (column 1, panel A, first row). Meanwhile, the RD coefficient of the same model with registered employment in the pre-application period (second row) is also negative, smaller (−3.8 percentage points) and statistically significant at the $p < 0.05$ level, which is consistent with the previous discussion regarding the balance in the pre-treatment outcomes between those eligible and those ineligible and with the evidence in Table A.4.3 and panel A in Fig. 3. Accounting for this pre-application difference, our DD-RD estimate of the effect of eligibility for AFAM on registered employment is a negative and statistically significant effect of 5.6 percentage points (third row). In terms of the mean outcome of the comparison group, this effect represents a proportional decrease of about 12% in rate of registered employment.

The regressions presented in panel B of Table 1 include controls for a series of predetermined covariates. The pre- and post-application RD estimates are statistically significant and similar in magnitude to those reported in panel A. The DD-RD coefficient for the impact on registered employment is slightly higher (6 rela-

Table 1
Effect of AFAM eligibility on registered employment.

	Estimates (1)	BW (2)	Comparison Mean (3)	%Δ w.r.t. (1) (4)	Obs. (5)
Panel A. No Covariates					
Elegible(RD Post)	−0.094	0.049	0.462	−20.348	213,720
	(0.020)			(4.398)	
	[0.000]			[0.000]	
Elegible(RD Pre)	−0.038	0.062	0.373	−10.130	454,323
	(0.017)			(4.567)	
	[0.040]			[0.040]	
Elegible(DD-RD)	−0.056	0.050	0.462	−12.205	214,248
	(0.020)			(4.388)	
	[0.013]			[0.013]	
Panel B. With Covariates					
Elegible(RD Post)	−0.090	0.052	0.463	−19.412	225,768
	(0.019)			(4.042)	
	[0.000]			[0.000]	
Elegible(RD Pre)	−0.030	0.068	0.375	−8.048	516,779
	(0.016)			(4.156)	
	[0.061]			[0.061]	
Elegible(DD-RD)	−0.060	0.052	0.463	−12.948	226,056
	(0.019)			(4.039)	
	[0.003]			[0.003]	

Notes: The sample corresponds to heads of households, and spouses of heads of households, aged 18 to 57 at the time of the AFAM application during the period January 2008-September 2010, from households within the eligibility score range [−0.257; +0.257]. The dataset corresponds to the *AFAM administrative records matched with the SSA registered employment records* for the pre-application (36 months) and post-application (24 months) period according to the SSA's administrative records (see Section 3.2 for a detailed description of the data). Panels A and B present the estimates from the RD –using either pre (row 1) or post (row 2) application data –and DD-RD (row 3) models in Eqs. (A.4.1, (2)) without and with socioeconomic covariates, respectively. Estimates from local linear regressions use a triangular kernel. The dependent variable is registered employment, measured as an indicator variable equal to 1 if the SSA records indicate that there are social security contributions from employment for the individual in a given calendar month, and zero otherwise. The covariates in the regressions in panel B include gender, head of household status, age, marital status, educational level (in 3 categories), the number of children aged 0–17 in the household, whether the household was enrolled in the PANES program, residence in Montevideo (Uruguay's capital), and whether the individual was employed. All regressions include time and date-of-application fixed-effects. Column (1) reports "conventional estimate" coefficients and household-clustered standard errors (curved parentheses), and p-value "bias-corrected estimates" (square parentheses) following Calonico et al. (2014). "BW" in Column (2) reports optimal bandwidths following the Calonico et al. (2014) procedure. "Comparison Mean" in Column (3) reports the average of the dependent variable for ineligible individuals within the score bandwidth. Column (4) reports the program's effect from column (1) as a percentage of the mean of the dependent variable for ineligible individuals at the cutoff (columns 3). Column (5) reports the total number of observations.

tive to 5.6 –third row in panels B and A respectively), and they are both significant in statistical and economic terms. The pattern of results in Table 1 is consistent with the hypothesis discussed in Section 2.4: we find that eligibility for a social assistance program such as AFAM induced decline in registered or formal employment.[31]

Finally, we can also exploit the longitudinal nature of the administrative data to study the effect of AFAM eligibility on the evolution of registered employment over time before and after application to the program. We start with panel A in Fig. 4, which gives a sense of the pattern of program participation after application. The figure presents the RD differences in the participation probability of eligible and ineligible groups since the household submitted its application, along with estimated confidence intervals.[32] The two vertical dashed lines indicate the semester when the household applied to the AFAM and the last semester of our data, respectively. As expected, the probability of program enrollment increases rapidly once the household applies and become eligible. In the application semester ($t = 0$), eligible households are 31.8 percentage points more likely than ineligible ones to be enrolled in the program. This value increases to 90.6 percentage points in the third semester after application.

Fig. 4 (panel B), in turn, plots the effect of the program on registered employment by semester, that is, the evolution of the RD estimates over time, with the baseline period given by each household's application date. The difference in registered employment between the two groups is not statistically significant in the semesters before application. In the application semester, when the difference in enrollment is still relatively low (31.8 percentage points), the difference in registered employment between the two groups is negative (3.2 percentage points) and significant at the $p < 0.05$ statistical level. The difference in registered employment increases as the difference in enrollment rates increases. In the first semester after application, with an enrollment difference of 60.2 percentage points, the difference in registered employment is 7.9 percentage points, and it falls further in semesters 2 and 3 to a level of 9.2 percentage points when the enrollment difference is 72.3 and 90.5 percentage points respectively. There is a slight fall in the difference in registered employment to 8 percentage points in the fourth semester, when the difference in program participation rates also starts to fall (all differences after the application semester are highly statistically significant).

5.2. Distributional effects

The previous analysis considered average effects among all adult applicants. To elucidate the range of responses to the AFAM program, we now examine the degree of heterogeneity of responses by subgroups of individuals. Specifically, we analyze the heterogeneity of AFAM's impact by formal labor market attachment.[33] As Perry et al. (2007) argued, there are two ways of explaining the presence of informal work in an economy. According to the "exit" view, workers volun-

[31] Except when noted, the regression results presented in the remainder of the paper are based on DD-RD estimates with covariates, because by controlling for pre-application differences they represent a more conservative estimate of the program's effects.

[32] While we report the RD difference for consistency with the bottom panel, inspection of Fig. 2 indicates that enrollment for the non-eligible households was very low, and the RD differences reflect mostly enrollment levels for the eligible.

[33] Responses to AFAM could also have varied as a function of other individuals'and households'observable characteristics. However, this does not seem to be the case. Appendix Table A.6.2 (Appendix A.6.2) presents the DD-RD estimates for different subgroups of the *Main Sample*. These estimates do not reveal any clear pattern of heterogeneous effects by type of individuals, with the exception of single female households, for whom the effect of AFAM is substantially larger than for other groups.

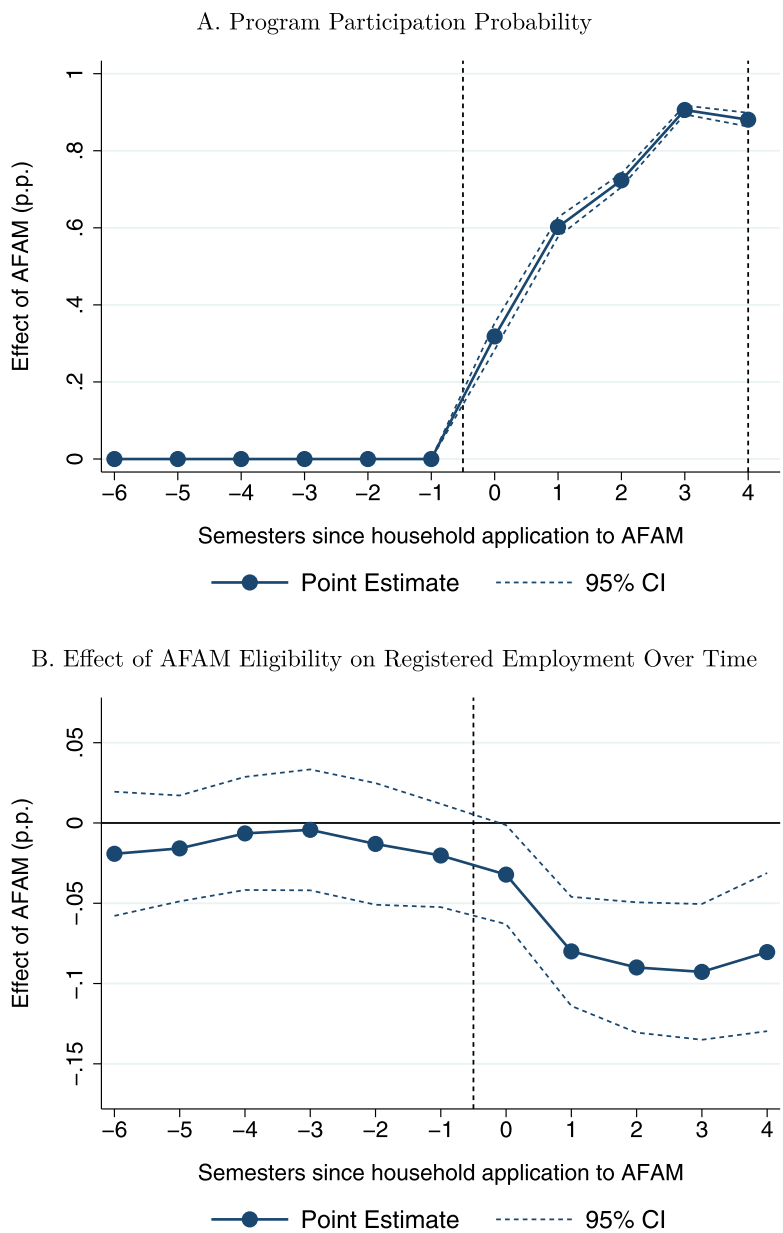

Fig. 4. Effect of AFAM Eligibility on Registered Employment Over Time. Enrollment and Registered Employment Estimates by Semester.*Notes*: The sample corresponds to heads of households, and spouses of heads of households, aged 18 to 57 at the time of AFAM application during the period January 2008-September 2010. The dataset corresponds to the *AFAM administrative records*matched with the *SSA registered employment records* for the pre-application (36 months) and post-application (24 months) period according to the SSA's administrative records (see Section 3.2 for a detailed description of the data). Each point (solid circle) in the plot represents the estimate of the RD effect of eligibility on the probability of participation in AFAM (panel A) or registered employment (panel B) for a specific semester from one to six semesters before application to AFAM, and up to four semesters after applying to the program. The estimates correspond to a local linear regression, estimated separately on either side of the eligibility cutoff. Regressions in panel B. include covariates as controls (see notes to Table 1). The dotted lines plots a 95% confidence interval based on standard errors clustered at the household level.

tarily choose to engage in informal employment because they place less value on the social insurance benefits tied to registered jobs relative to their costs. That is, they seek to avoid the associated payroll taxes and contributions because, in the context of weak enforcement of tax and labor regulations, informal employment results in better net-pay from a private cost-benefit calculation. According to the "exclusion" view, by contrast, workers are informal because formal jobs are harder to find (e.g. because search costs are higher for formal jobs). These two views can be considered complementary.[34] In any case, with or without segmented labor markets, theory predicts differential effects among individuals with different propensities to work as formal employees –i.e., those closer (marginal) or further away (infra-marginal) from the registered/not registered employment margin of choice (Galiani and Weinschelbaum, 2012). Even if all AFAM-eligible individuals face the same incentives from the program, their outside options might differ (for instance, given their levels of education), which is what our proxy for the probability of being formally employed (described below) captures. The negative effect on registered employment should be higher among individuals with higher propensities to be employed as formal workers relative to potential beneficiaries with lower propensities to be formally employed.

[34] Early versions of the exclusion view considered formal and informal sectors as segmented –i.e., the presence of dual labor markets (Fields, 2009). However, several studies using longitudinal data challenged this view by showing that workers often transit between formal and informal jobs in Latin America (Maloney, 2004).

We combine a broad set of background characteristics into a single predicted registered employment index as a proxy for the propensity to work formally in the absence of AFAM. We estimate a regression model with registered employment as a function of our standard set of covariates only for the group of ineligible individuals (Appendix Table A.6.1 reports the results from this auxiliary regression).We then usethe estimated coefficients to generate predicted values of the probability to work as a registered employee for all individuals in the sample (i.e., also including those eligible for AFAM).We then divide our sample into three roughly equally-sized groups based on this score. Finally, we estimate the effects of the program for each of the three subgroups separately. This approach follows what Abadie et al. (2018) call the "endogenously stratified" group method.[35]

We perform the distributive analysis considering both the full sample of individuals and the group of single mothers. Single mothers represent a large share of our full sample (about 43%) of AFAM beneficiaries.Moreover, the literature on welfare programs' effects has documented stronger behavioral responses for this group compared to the rest of the population (Moffitt, 2002; Chan and Moffitt, 2018). Indeed, we find that single mothers is the group that responds most strongly to AFAM in our sample (Table A.6.2).

Table 2 reports the estimates of the effect of AFAM eligibility on registered employment for these three sub-groups, both for all beneficiaries (panel A) and for single mothers (panel B). The table follows the column layout of Table 1, but presents DD-RD estimates only.[36] Column (3) presents the average of the dependent variable, registered employment, for ineligible individuals in each subgroup at the cutoff. The average for the full sample is 46.3%, with a slightly lower level (45.2%) for single mothers. As expected, individuals in the group with the lowest propensity to work formally exhibit a low level of registered employment—13.4%—in the post-application period (panel A), with an even lower level for single mothers (12.3%, panel B). The levels are substantially higher for individuals in the middle and high propensity groups –30.7% and 78.7%–, with lower levels for single mothers (25.0% and 71.5%).

The effects of the program on registered employment are presented in Column (1) of Table 2. The effect for the full sample is 6 points percentage negative (panel A, as in the panel with covariates in Table 1), and substantially higher in absolute value for single mothers –8.7 percentage points (panel B). Both effects are highly statistically significant. The DD-RD estimate for the group with a low propensity to work formally is still negative but lower in magnitude and not statistically significant at conventional levels for the full sample (2.5 percentage points), and positive but not statistically significant for single women with a low propensity. For the group with a middle propensity, the effect is negative and substantially higher for the full sample (−8.9 percentage points) and for single mothers (−12.4 percentage points). Finally, it is also negative for the group with a high propensity to work formally at −4.7 percentage points for the full sample and −9.8 percentage points for single mothers, although only the latter is statistically significant at conventional levels. The effects as a proportion of the comparison mean differ substantially: they represent about −13% for the full sample and −19.3% for single mothers, but they are much larger for the middle-propensity group (−29% and −49.5%, respectively) than for the high propensity group (−5.9% and −13.6%). The strongest behavioral response to the program, both in absolute and in relative terms, is observed for the group with a middle propensity to work formally, whereas the statistically insignificant effects for the low propensity groups (for the full sample and for single mothers) indicate that this group is much less responsive to the incentives introduced by the program.[37]

These results are consistent with the fact that individuals in these groups probably face different incentives and have different opportunities in the labor market. On the one hand, those with a low propensity to work formally may not respond much to the financial incentives of the program because they have limited opportunities to work as registered employees to begin with. On the other hand, individuals with a high propensity to work formally are less affected by the program's disincentives for registered employment, because most of them are likely to work formally regardless. The group with the middle propensity to be formally employed seems to be the one closest to the margin of choice between formal and informal employment, and thus individuals in this group are those who react most strongly to the new incentives. The presence of heterogeneous treatment effects in the dimension of propensity to work formally accords with the evidence documented in Gerard and Gonzaga (forthcoming), who find heterogeneity across labor markets with different formal employment rates in Brazil. Our evidence complements these results by analyzing heterogeneity between groups within the same labor market.

5.3. The implied elasticity of participation in registered employment

The results presented thus far indicate that individuals eligible for AFAM respond to the program's financial incentives by reducing their participation in registered employment. In this section, we establish the magnitude of these responses in relation to the size of the program's financial incentives. We use these results to establish a more general parameter, the aggregate elasticity of participation in registered employment to net-of-(registered employment) participation tax rate –that is, the percentage change in formal employment for each percentage point change in the net average tax rate.[38] This parameter captures movements into and out of formal employment as a consequence of the tax and transfer schedule.

We compute the aggregate elasticity of participation in registered employment ϵ_R in the context of our DD-RD analysis by closely following the procedure developed by Kostol and Mogstad (2014). Specifically we define ϵ_R as:

$$\epsilon_R = \frac{1 - ptr_{ineligible}}{R_{ineligible}} \cdot \frac{\triangle R}{\triangle(1 - ptr)} \tag{3}$$

where $(1 - ptr_{ineligible})$ denotes the mean of the net-of-participation tax rate in registered employment for the ineligible group, $\triangle(1 - ptr)$ is the difference in the net-of-participation tax rate between eligible and ineligible individuals, $R_{ineligible}$ represents the registered employment rate for AFAM ineligible individuals, and $\triangle R$ denotes the difference in registered employment between eligible and ineligible individuals. It should be noted that the ptrcorresponds to the effective average tax rate, which includes both payroll taxes on formal earnings and the loss of the program's ben-

[35] Alternatively, we can use the individual history of registered employment in the pre-application period as a proxy for the propensity to be a formal worker. We define it as the proportion of months that an individual was a registered worker in SSA records during the 36-month pre-application period. The estimates remain qualitatively similar when we implement this alternative for group stratification. These results are available upon request.

[36] Appendix Tables A.6.3 and A.6.4 report the RD estimates underlying our DD-RD results using pre- and post-application data, respectively.

[37] The difference between the coefficient for the middle propensity group and that for the other two groups is statistically significant for single mothers, although only the difference between the middle- and low-propensity groups is significant for the full sample.

[38] As discussed in Section 2.4, AFAM's income test eligibility rule introduces a discrete change in potential beneficiaries' budget sets. In the context of such a tax and transfer schedule, the participation tax rate is generally considered more relevant to behavioral responses than the marginal tax rate (Eissa et al., 2006; Eissa et al., 2008).

Table 2
Effect of AFAM eligibility on registered employment, by propensity to be a registered employee, full sample and single mothers.

	Estimates (DD-RD)	BW	Comparison Mean	%Δ w.r.t. (1)	Obs.
	(1)	(2)	(3)	(4)	(5)
Panel A. Full Sample					
Average	−0.060	0.052	0.463	−12.948	226,056
	(0.019)			(4.039)	
	[0.003]			[0.003]	
Low	−0.025	0.075	0.134	−18.572	113,568
	(0.017)			(12.478)	
	[0.182]			[0.182]	
Medium	−0.089	0.073	0.307	−29.062	118,056
	(0.025)			(8.239)	
	[0.000]			[0.000]	
High	−0.047	0.051	0.787	−5.926	86,496
	(0.026)			(3.342)	
	[0.104]			[0.104]	
Panel B. Single Mothers					
Average	−0.087	0.064	0.452	−19.265	147,672
	(0.023)			(5.119)	
	[0.000]			[0.000]	
Low	0.039	0.038	0.123	31.771	18,408
	(0.030)			(24.696)	
	[0.118]			[0.118]	
Medium	−0.124	0.056	0.250	−49.518	38,160
	(0.036)			(14.236)	
	[0.001]			[0.001]	
High	−0.098	0.052	0.715	−13.649	47,088
	(0.039)			(5.407)	
	[0.019]			[0.019]	

Notes: The sample corresponds to heads of households, and spouses of heads of households, aged 18 to 57 at the time of the AFAM application during the period January 2008-September 2010, from households within the eligibility score range [−0.257; +0.257]. The dataset corresponds to the *AFAM administrative records* matched with the *SSA registered employment records* for the pre-application (36 months) and post-application (24 months) period according to the SSA's administrative records (see Section 3.2 for a detailed description of the data). Each row presents the estimates from the DD-RD model in Eq. (2) with covariates at time of application to the program for the corresponding subgroup, as in the notes to Table 1. Estimates from local linear regressions use a triangular kernel. The dependent variable is registered employment, measured as an indicator variable equal to 1 if the SSA records indicate that there are social security contributions from employment for the individual in a given calendar month, and zero otherwise. Panels A and B present the regression estimates for the full sample and the sample of single mothers, respectively. The first row in each panel reports the average estimate for each sample, while rows 2 to 4 report estimates for the subgroups of individuals with Low, Medium, and High propensity to work as a registered employee, respectively. See Section 5.2 for a detailed explanation of the procedure used to determine these probabilities. Column (1) reports "conventional estimate" coefficients and household-clustered standard errors (curved parentheses), and p-value "bias-corrected estimates" (square parentheses) following Calonico et al. (2014). "BW" in Column (2) reports optimal bandwidths following the Calonico et al. (2014) procedure. "Comparison Mean" in Column (3) reports the average of the dependent variable for ineligible individuals within the score bandwidth. Column (4) reports the program's effect from column (1) as a percentage of the mean of the dependent variable for ineligible individuals at the cutoff (column 3). Column (5) reports the total number of observations.

efit following entry into registered employment (when the resulting earnings exceed the program's income test threshold).

We obtain estimates of $R_{ineligible}$ as the average registered employment rate for ineligible individuals at the cutoff point, while $\triangle R$ corresponds to the DD-RD estimates with controls discussed in the previous section. The main challenge to compute this elasticity is to find a measure of the $ptr_{ineligible}$–i.e., the implied *ptr* for the difference between disposable income when formally employed (subtracting payroll taxes) and when not formally employed (including the AFAM cash transfer when relevant).This requires measures of individual earnings and of net-of-tax transfers in both employment states. Since we do not have information on earnings in the SSA records, we impute them for formal and informal employment based on Uruguay's ECH household survey for the period 2008–2012 (i.e., the period covered by our SSA data). As expected, ineligible individuals have higher average earnings than eligible individuals. We also compute individual net taxes and transfers based on Uruguay's tax code and on the AFAM transfer schedule, which we apply to each individual and her household characteristics. Importantly, the simulated schedule accounts for AFAM benefit loss when earnings from registered earnings are above the program's eligibility income threshold. Based on these inputs, we estimate the disposable income and compute the *ptr* for every individual in our database. Finally, we compute $\triangle(1 - ptr)$ as the weighted difference in the net participation tax rate between eligible and ineligible individuals. The weights are given by the conditional density of the registered earnings of ineligible individuals who work as registered employees.[39]

Table 3 presents the estimated elasticities for the three subgroups of propensity to be a registered employee for individuals in the full sample (panel A) and for single mothers (panel B). Each row reports our estimates of the components of Eq. (3), and the final row in each panel presents the resulting elasticity and its confidence interval.

The change in registered employment, our proxy for $\triangle R$, is presented for each subgroup in the first row of each panel, and corresponds to the estimates reported in Table 2. The change for the full sample is −6 percentage points, and −8.7 for single mothers, with higher effects for the group with medium propensity to be a formal worker. The average registered employment level for ineligible individuals at the cutoff, our proxy for *R*, is 46.3% for the full sample and 45.7% for ineligible single mothers. These levels vary accordingly by propensity to be formally employed: the registered employment rates are 13.4%, 30.7% and 78.7% for those with low, medium and high propensities, respectively, to work as registered employees in the full sample, and 12.3%, 25.0%, and 71.5% percent for the corresponding subgroups of single mothers.

[39] Appendix A.3 details the earnings imputation procedure, the computation of the weights, and other details on the derivation of this elasticity in our context. Appendix Fig. A.3.1 illustrates the net-of-tax*ptr*,i.e. 1-*ptr*, and the density of simulated registered employment earnings by eligibility status for the year 2011.

Table 3
Elasticity of participation in registered employment.

	Average (1)	Propensity to be formal: Low (2)	Medium (3)	High (4)
Panel A. Full Sample				
Δ Registered Employment	−0.060	−0.025	−0.089	−0.047
	(0.019)	(0.017)	(0.025)	(0.026)
Registered Employment (Ineligible)	0.463	0.134	0.307	0.787
	(0.499)	(0.340)	(0.461)	(0.409)
Δ Net-of-Participation Tax Rate	−0.062	−0.055	−0.065	−0.071
	(0.001)	(0.001)	(0.001)	(0.001)
Net-of-Participation Tax Rate (Ineligible)	0.375	0.394	0.368	0.366
	(0.026)	(0.023)	(0.033)	(0.026)
Elasticity (ϵ_R)	0.784	1.336	1.641	0.308
	[0.612–0.912]	[0.987;1.812]	[1.235;1.911]	[0.224–0.424]
Panel B. Single Mothers				
Δ Registered Employment	−0.087	0.039	−0.124	−0.098
	(0.023)	(0.030)	(0.035)	(0.039)
Registered Employment (Ineligible)	0.457	0.123	0.25	0.715
	(0.498)	(0.328)	(0.433)	(0.451)
Δ Net-of-Participation Tax Rate	−0.055	−0.047	−0.053	−0.052
	(0.001)	(0.002)	(0.001)	(0.001)
Net-of-Participation Tax Rate (Ineligible)	0.372	0.358	0.370	0.370
	(0.060)	(0.057)	(0.060)	(0.061)
Elasticity (ϵ_R)	1.288	−2.415	3.463	0.975
	[1.080;1.428]	[−3.684;−1.362]	[2.688;3.989]	[0.776;−1.230]

Notes: The sample corresponds to heads of households, and spouses of heads of households, aged 18 to 57 at the time of the AFAM application during the period January 2008-September 2010, from households within an eligibility score range[−0.257; +0.257]. The dataset corresponds to the *AFAM administrative records* matched with the *SSA registered employment records* for the pre-application (36 months) and post-application (24 months) period according to the SSA's administrative records, and to the ECH for the period 2008–2012 (see Section 3.1 for a description of the data). The elasticity formula is given in Eq. (3) and discussed in Section 5.3 and A.3. Panels A and B report the estimates for the full sample and the sample of single mothers, respectively. In the rows, "△ Registered Employment" represents the differences in registered employment between eligible and ineligible individuals as given by the DD-RD estimates in Table 2. "Registered Employment (Ineligible)" denotes the registered employment rate for ineligible individuals at the cutoff (the comparison group). The "Net-of-Participation Tax Rate (Ineligible)" is given by Eq. A.3.7, and computed as the mean value for ineligible individuals (see Section A.3 for a detailed explanation). The differences in this tax between eligible and ineligible individuals are given by Eq. A.3.6. Column (1), "Average", refers to the estimates for the whole sample. Columns (2) to (4) report the estimates for the subgroups of individuals with Low, Medium, and High propensity to work as a registered employee, respectively. Confidence intervals on the estimated elasticity (square parentheses) are computed using a bootstrap method.

The third row in each panel presents the average change in the net-of *ptr*, given by the loss of the entire AFAM cash transfer at the income-eligibility threshold. While the implicit tax rate on the benefit is 100% at the threshold, we average out over the whole distribution to obtain the change in the average tax rate. This rate is −6.2% for the full sample and −5.5% for single mothers. The rate ranges from −5.5% to −7.1% for the three subgroups of the full sample, whereas it is more homogeneous for the three subgroups of interest among single mothers –from −4.7% to −5.3%. Finally, the forth row in each panel shows that the net-of-participation tax rate in formal employment for ineligible individuals is 37.5% for the full sample and 37.2% for single mothers, with little variation between the subgroups (36.6–39.4% for subgroups of the full sample, and 35.8–37.0% for single mothers).

The bottom row in each panel of Table 3 presents the resulting estimates for the registered employment elasticity. For the full sample, it stands at 0.784, and it is substantially higher (1.288) for single mothers. These figures mean that a reduction of 1% in the net-of-tax share of income that individuals can keep reduces registered employment by about 0.8% for the full sample, and by about 1.3% for single mothers.

Consistent with the previous results regarding the heterogeneity of AFAM's impact, we find a substantially larger elasticity for the group with a medium propensity to be a registered employee: 1.641 for the full sample, and 3.463 for single mothers, reflecting the fact that these are probably the individuals closest to the formal/informal margin. The elasticities are lower for the group with a higher propensity to be formal (0.308 for the full sample and 0.975 for single mothers), reflecting the fact that individuals in this group are likely to work as registered employees in any case. Finally, the elasticities for the group with a low propensity to be formally employed are large and positive for the full sample (1.336), due to the low rate of registered employment, and large and negative for single mothers (−2.415), due to the positive estimated impact on registered employment for this subsample. It should be noted that even if the confidence intervals for the last two elasticities do not include zero, the reduced-form effects (−0.025 and 0.039) are not statistically significant at standard levels (i.e., p-values are 18.2% and 11.8%, respectively).

Our results for Uruguay indicate that individuals in developing countries make decisions about entering formal employment in response to the incentives implied by the tax and transfer schedule they face. Moreover, these estimates accord with those found in developed countries. For instance, the corresponding elasticity for single mothers in the USA derived from variation in the generosity of the AFDC program is about 0.7 (based on estimates from Meyer and Rosenbaum, 2001), compared to our estimate of about 1.3 for this group. However, in a context like ours with high informality, our estimates reflect the response in terms of non-participation and also in terms of movements between formal and informal employment. We find substantial heterogeneity along this dimension: the elasticity is much higher (about 3.5) for single mothers on the margin of being formal employees, with lower levels for those with either very high or very low propensities to work formally. The following section attempts to shed some light on how our results concerning lower registered employment map onto these two alternative outcomes.

5.4. Decomposing the effects of the program: informality and non-employment

The main analysis in the previous section was based on the *Main Sample*, a longitudinal dataset of formal work histories from *SSA registered employment records*. As discussed above, this high-

quality data source has a limitation in terms of the outcomes of interest: while we can establish the effects of the program on changes in registered employment, we cannot separate the responses in terms of inactivity or non-registered employment – i.e., we cannot know whether the null registered employment status in the administrative data represents a period in which the individual unemployed or, alternatively, a period of informal employment. This is an important distinction for understanding how programs such as AFAM interact with a labor market featuring high informality.

Our *Follow-Up Survey* provides information to complement the benchmark specifications for registered employment, allowing us to complete the anatomy of responses to the AFAM's financial incentives by establishing the comparative magnitude of responses along the non-employment and informal work margins. The analysis, however, has some clear limitations. Unlike the *Main Sample*, the *Follow-Up Sample* is a cross-sectional dataset, and a relatively small sample at that. We thus cannot compute DD-RD estimates and must rely on standard RD, which may be biased because we cannot correct for factors such as the baseline differences in the main outcome identified in Section 4.[40]

Moreover, we do not have enough statistical power to distinguish between the full sample and single mothers, and thus we only present results for the former. For these reasons, we interpret the following results as merely suggestive.

Table 4 presents the baseline estimated RD effects (i.e., Eq. (2) without the interaction terms) for this sample on three mutually exclusive labor market outcomes: registered employment, non-employment and unregistered or informal employment. For comparison purposes, we present two sets of results. First, in column (1) we present the RD estimates by using the self-reported outcome from the *Follow-Up Survey*. Second, in column (2), we present our preferred set of RD results based on the sub-sample of individuals whose self-reported registered employment coincided with their registered employment status in the SSA records at the time of the survey's interview.[41] We denote this group the "consistent sample".

The results in Table 4 indicate that the negative effect on registered employment is similar in size for the different estimations, and all estimates are statistically significant at the $p < 0.01$ level (row 1). Our preferred estimates (those based on the consistent sample) indicate a statistically significant effect on registered employment of about 10 percentage points, which might be biased upwards because we are not controlling for baseline differences in the outcome. The following two rows present the main result from this exercise. Focusing on the consistent sample (column 2) as a benchmark, the 10 percentage points reduction in registered employment induced by AFAM corresponded to an increase in non-employment of about 5 percentage points, and a similar increase in informal employment. However, only the coefficient for non-employment is statistically significant at the usual statistical levels. The p-value for the test that both coefficients are equal to zero is 0.104 (row 3), and we cannot reject the null hypothesis that they are both equal (p-value of 0.975, in row 4).

In the discussion of the expected effects of the program, we proposed that the cash transfer could cause an increase in non-employment or a shift towards informal employment. Even with the study's limitations, the suggestive evidence in this section indicates that both effects are probably present in the context of AFAM.

Table 4
Effect of AFAM eligibility on different margins of participation.

	Self-Reported Follow-Up Survey (FUS) (1)	SSA Records = FUS (2)	Comparison Mean (3)
1. Registered Employment	−0.109	−0.100	0.548
	(0.041)	(0.042)	
	[0.003]	[0.002]	
2. Non-Employment	0.050	0.051	0.189
	(0.036)	(0.037)	
	[0.056]	[0.043]	
3. Informal Employment	0.059	0.049	0.263
	(0.043)	(0.044)	
	[0.218]	[0.209]	
Joint p-value (2 & 3 = 0)	0.065	0.104	
Joint p-value (2 = 3)	0.911	0.975	
Observations	2,403	2157	

Notes: The sample corresponds to heads of households, and spouses of heads of households, aged 18 to 57 at the time of the AFAM application during the period January 2008-September 2010. This sample corresponds to the subset of individuals who were interviewed for the program's follow-up survey during theperiod September 2011-February 2013. The survey's sample was drawn from households within the eligibility score range $[-0.0426; +0.0727]$ (see Section 3.1 for more details). Each row presents the estimates from the DD-RD model in Eq. (2) excluding the $Post_t$ terms and the corresponding interactions and with covariates at the time of application to the program, as the notes to Table 1 explain. Estimates from local linear regressions use a triangular kernel. In the first row, the dependent variable is the registered employment status (one if she is working as a registered employee, zero otherwise). In the second row, the dependent variable is non-employment, which is equal to on1 if the individual is not working, and equal to zero if she is working (as a registered or as a non-registered employee). In the third row, the dependent variable is informal employment at the time of the interview, which is equal to 1 if the individual is working as an informal (or non-registered) employee, and zero otherwise (i.e., if she is working as a registered employee or not working). The fourth (fifth) row presents the p-value of the statistical test that the eligible coefficients in the "Non-Employment" and "Informal Employment" regressions are jointly equal to zero (equal). Column (1), "Self-Reported Follow-Up Survey (FUS)", presents results for individuals in the *Follow-Up Sample* using the self-reported information they provided at the interview to measure the outcome variables. Column (2), "SSA Records = FUS," reports results based on the subsample of individuals who reported being employed as registered employees in a manner consistent with the SSA records for the same period when they were interviewed. We report "conventional estimate" coefficients and household-clustered standard errors (curved parentheses), and p-value "bias-corrected estimates" (square parentheses) following Calonico et al. (2014). "Comparison Mean" in Column (4) reports the average of the dependent variable for ineligible individuals close to the cutoff,i.e. within the $[-0.0426; 0]$ score bandwidth.

6. Efficiency cost implications of the AFAM program

The empirical analysis performed thus far has shown that the AFAM program might generate behavioral distortions in the labor market by reducing participation in registered employment. While this might be an important policy question, the ultimate relevant outcome is the welfare impact of the program and its incentives. We follow Hendren's, (2016) approach to translate the estimated causal effects of a public policy change to a measure that captures its efficiency cost implications.[42] The key statistic from this approach is the marginal value of public funds (*MVPF*).

[40] Appendix Table A.4.3 presents the covariate balance tests for adults in the *Follow-Up Sample* and Figs. A.5.4–A.5.6 the depict the corresponding RD plots.

[41] Appendix Tables A.6.5–A.6.7 present cross-tabulations of self-reported and SSA-recorded registered employment. The classification errors in the registered employment variable are about 10 percent, and are similar in magnitude when conditioning by eligibility status (10% for eligible individuals vs. 8% for the ineligible). Fig. A.5.7, in turn, indicates that the RD difference in these reporting errors is balanced by eligibility condition.

[42] Hendren (2016) shows that the government is the only distortion between private prices and social costs. Thereby, the causal (behavioral) response to a policy change regarding the government's budget is the only statistic required for welfare calculations.

This measure is given by the ratio of marginal benefits to marginal costs of the extra funds spent on the policy. Specifically, the marginal benefit is defined by the beneficiaries' marginal willingness to pay (*WTP*) for the increase in expenditure out of their own income to the cost to the government of the additional funds spent per beneficiary. This net cost includes both the increase in government expenditure due to the policy in the absence of any behavioral change (denoted as the "*mechanical cost*") and all other impacts on the government budget of behavioral responses to the policy (denoted "*fiscal externalities*"). Thus, the *MVPF*of a policy *j* is defined as*:*

$$MVPF^j = \frac{WTP^j}{Net\ Cost^j} \tag{4}$$

In our setting, the object of interest is the $MVPF^{AFAM}$ of the extra local currency unit (UYU) spent on the AFAM program. If we make the plausible assumption that AFAM's beneficiaries value each UYU in monetary transfers as cash earnings, the envelope theorem suggests that they are willing to pay for the mechanical cost of the AFAM benefit –i.e., UYU 1 transfer is valued at UYU 1.[43] Normalizing the *MVPF* in Eq. (4) by the mechanical cost of AFAM, the *MVPF* is valued at UYU 1 in mechanical cost per program beneficiary. Thus, the MVPF of an extra UYU 1 spent in the AFAM cash transfer is:

$$MVPF^{AFAM} = \frac{1}{1 + FE^{AFAM}} \tag{5}$$

where FE^{AFAM} describes the effect of the behavioral response to AFAM on the government's budget (i.e., the fiscal externality) per UYU increase in the mechanical cost per AFAM beneficiary. Importantly, $MVPF^{AFAM}$ captures the relative gain for beneficiaries of receiving the AFAM transfer vs. receiving a lump-sum from the government, holding the government budget constant. Thus, in our setting, $MVPF^{AFAM}$ measures the efficiency cost of the AFAM program. In what follows, we describe how we estimate $MVPF^{AFAM}$.

WTP: The willingness to pay for the AFAM transfer is given by the marginal benefit of the program per UYU 1 of mechanical cost, and is equal to UYU 1 (numerator in Eq. (5)). Intuitively, this means that increasing the cash benefit by UYU 1 is valued at UYU 1 for the average AFAM recipient. We do not need the estimated causal effects for this estimate.

Net Cost: This component represents the marginal cost of increasing the AFAM cash transfer by UYU 1, and it is given by $1 + FE^{AFAM}$ (the denominator in Eq. (5)), which adds the fiscal externality per UYU 1 induced by AFAM to the mechanical cost (UYU 1). In our calculations, we only add the fiscal externality resulting from the reduction in adults' registered employment as a response to AFAM during the period under study –i.e., two years after the beneficiaries enrolled in the program. Thus, the estimated *MVPF* does not account for long-run responses in the labor market nor for spillover effects on children (for instance, in terms of increased education given the program's conditionalities). We discuss below the implications of including these effects in our *MVPF*estimate.

The impact of the decline in registered employment in response to AFAM has two fiscal externality components that we incorporate in FE^{AFAM}. First, the behavioral response reduces tax revenue –i.e., payroll taxes –collected by the authorities because of the fall in registered employment earnings. Second, AFAM's effects on registered employment also imply higher government expenditures, because in the absence of the program some of the individuals who would be registered workers would also not draw the AFAM benefit. Note that some of the individuals who become informal would have been AFAM beneficiaries even when formally employed, because their earnings would be below the eligibility threshold. Among those who changed their behavior, we should include in our accounting only the fraction who received AFAM payments because of their behavioral responses.

We need to make a series of assumptions to estimate these components. First, we assume that the decline in registered employment due to AFAM does not create fiscal externalities for other tax bases. Although capital income taxation is not an issue for AFAM beneficiaries, this assumption rules out fiscal externalities with respect to other important tax bases in developing countries, such as consumption. Second, we also rule out fiscal externalities of the AFAM program with respect to take-up of other social assistance programs. Third, as in our previous calculations, we assume that the AFAM population is only subject to the payroll tax, which amounts to about 32% of gross earnings (the sum of the employee and employer taxes) and is paid by the employer. Fourth, we use the distribution of *imputed* formal earnings (used for calculations in Section 5.3) for eligible individuals to compute the average counterfactual (gross) formal earnings of AFAM beneficiaries who reduced their level of registered employment because of the program and the proportion of individuals that would be above or below the eligibility threshold in this counterfactual.[44] The earnings distribution is based on imputation procedure developed to compute elasticities in Section 5.3 (see also Appendix A.3). Finally, we assume that the only mechanical cost to the government for providing the program to the beneficiary household is the cash transfer –that is, we do not factor in the program's administration and monitoring costs.

For the purposes of simplification and of comparability with previous studies, the following calculations illustrate the case of single mothers. Table 5 reports main inputs for the calculations and our results. Column 1 shows results for the average single mother, while columns 2 to 4 report the estimates based on breaking down the sample of single mothers into the three groups according to their propensity to be registered employees, as described in the previous sections.

Row A in Table 5 reports the mechanical cost calculations, which correspond to the average monthly benefit of AFAM per single mother household. We apply the benefit determination rule given by Eq. (1) (Section 2.3) to compute the AFAM benefit for the average single mother household and for the three subgroups. The average number of children per household is similar between groups, which explains the very similar values of the benefit between groups.

Row B, in turn, presents the change in payroll tax revenue as a consequence of AFAM-induced behavioral responses. We use the DD-RD estimates of the program's effect on registered employment for the average eligible single mother and for the three subgroups (rows 5 to 8 in Table 2). Second, we apply a 32% tax rate on earnings. Third, we need an estimate of the average counterfactual earnings from registered employment –i.e., the gross registered employment earnings that single mothers would have accrued were they ineligible for AFAM. We proceed as follows: The DD-RD estimates indicate an average 8.7 percentage point decline in

[43] For AFAM beneficiaries who do not change their registered employment behavior in response to the program, the 1 UYU transfer is valued at 1 UYU. For those who modified their behavior because of the program (e.g., they exited registered employment to remain eligible), the envelope theorem implies that, since they already are optimizing, they should be indifferent to the policy change. Their willingness to pay is thus not counted for these computations –we only take into account the willingness to pay of infra-marginal individuals.

[44] Alternatively, we could have used the imputed distribution for ineligible individuals as a counterfactual to compute the statistics needed for MVPF calculations. The results are qualitatively the same and quantitatively very similar under this alternative assumption. For instance, the MVPF for all single mothers in this alternative is 0.55, similar to the value of 0.61 reported in Table 5. The full set of alternative results analogous to those in Table 5 is available upon request.

Table 5
Estimates of AFAM's Marginal Value of Public Funds (MVPF) for single mothers.

Component	Propensity to be formal			
	Average (1)	Low (2)	Medium (3)	High (4)
A – Mechanical Cost (UYU)	*1086*	*1080*	*1080*	*1126*
Average monthly AFAM benefit (UYU)	1086	1080	1080	1126
B – Change in Tax Revenue (UYU)	*592.7*	*– 956.7*	*1534.4*	*448.3*
Registered Employment Change (%)	19.3	−31.7	49.5	13.6
Payroll Tax Rate on Gross Earnings (%)	32	32	32	32
Gross Earnings of Single Mothers (UYU)	9597	9431	9687	10,302
C – Loss of Budget Saving (UYU)	*115.9*	*−171.2*	*304.2*	*105.9*
Average monthly AFAM benefit (UYU)	1086	1080	1080	1125
Registered Employment Change (%)	19.3	−31.7	49.5	13.6
% with Formal Earnings > Threshold	55.3	50.0	56.9	69.2
D – Fiscal Externality (B + C)/A	*0.65*	*−1.04*	*1.70*	*0.49*
MVPF (1/1 + D)	0.61	∞	0.37	0.67

Notes: Estimates of "Average Monthly AFAM Benefit" are obtained from the AFAM administrative data. "Registered Employment Change" is taken from Table 2 Estimates of "Gross Earnings of Single Mothers" and "% with Formal Income > Threshold" are obtained from Section A.3.3. The "MVPF" estimates follow Eq. (5).

registered employment, a 19.3% decrease with respect to the average employment levels of the ineligible single mothers around the cutoff point (column 1). We compute gross counterfactual earnings from registered employment of UYU 9597 for an average single mother beneficiary in 2011. A 32% tax rate on earnings implies that the net change in tax collected from an individual due to the AFAM-induced reduction in registered employment is equal to UYU 592.7 $(0.193 \times 9597 \times 0.32)$. Similarly, we compute the change in payroll tax revenue for single mothers according to their propensity to be formally employed (columns 2 to 4). It should be noted that the DD-RD estimate for single mothers with a low propensity to be formally employed is positive, in contrast to the negative estimate for the average of the group and for the full sample. While the coefficient is not statistically significant at standard levels, we report the results for this group for completeness.

Row C in Table 5 reports the calculations for the loss of potential government budget savings. The average monthly AFAM benefit for eligible single mothers is UYU 1086, and this represents the unitary cost of the program for this group of beneficiaries (column 1). Based on the distribution of imputed registered employment earnings, we estimate that 55.3% of single mothers would have had earnings from registered employment above the income-eligibility threshold. Recall also that we estimate a 19.3% decline in registered employment for single mothers. Therefore, the percentage of single mothers who receive AFAM benefits because of their program-induced decrease in registered employment is $10.7 = (0.553 \times 0.193 \times 100)$. The government thus foregoes average savings of about UYU $115.9 = (1086 \times 0.107)$ per single mother due to the beneficiaries who would have been engaged in registered employment in the absence of a behavioral response to the program and, thus, would not have drawn the benefit because they would not have passed the income test. We proceed analogously for the three subgroups, noting that for those with a low propensity to be a registered employee AFAM represents a revenue increase, given the positive estimate of the program's effect on registered employment for this group.

The average total cost of the AFAM program per single mother is thus UYU 1794.6 –i.e. the sum of the mechanical cost (UYU 1086, row A) and behavioral responses (UYU 592.7 plus 115.9, rows B and C, respectively). This implies a total fiscal externality of UYU 0.65 $([592.7 + 115.9]/1086)$ as reported in row D. About 84% (0.55/0.65) of the total fiscal externality corresponds to the loss of tax revenue due to the decline in registered employment, while roughly 16% (0.11/0.65) can be attributed to the loss of potential government budget savings. Columns 2 to 4 in row D show the fiscal externality estimates for single mothers according to their propensity to be formally employed.

Finally, the marginal value of public funds for the AFAM program for single mothers, $MVPF^{AFAM}$, is UYU 0.61 $(1/[1 + 0.65])$. An MVPF below 1 indicates that individuals would prefer a UYU 1 lump-sum transfer as opposed to UYU 1 in AFAM benefits. Our result implies that for every UYU 1 spent by the government on AFAM benefits, AFAM beneficiaries would be willing to pay UYU 0.56. This figure is higher –0.67 –for the subgroup of single mothers with a high propensity to be a registered employee and lower –0.37 –for the group with a medium propensity. Following Hendren and Sprung-Keyser (2020), we assign an infinite MVPF value to the group with a low propensity to be a registered employee, because this group exhibits a positive willingness to pay for the program but a negative net cost for the government. While this finding implies that the policy "pays for itself" for this group, the effect on registered employment for this group is not statistically significant.

This simple MVPF calculation provides an estimate of the order of magnitude of the efficiency cost implications of the program's effects on registered employment. However, our simple estimates abstract from some potentially important additional effects of the policy. First, our MVPF calculation ignores potential effects of AFAM on beneficiaries' children. For instance, the program's conditionalities could induce an increase in health check-ups and in school attendance, in turn generating an improvement in their educational, health and labor market outcomes in the long run. An increase in the earnings of beneficiaries' children represents a positive externality that would increase the MVPF with respect to our baseline calculation. While this scenario is plausible, the available preliminary evidence indicates that AFAM did not, in fact, induce a significant increase in health checkups or in school attendance (Bergolo et al., 2016). Moreover, while several studies of other developing countries find positive long-term impacts of CCTs on children's human capital accumulation, they fail to find higher lifetime earnings (Millan et al., 2019). Second, our MVPF calculation relies on a short-term impact of AFAM on the registered employment levels of adult beneficiaries. If the cash transfer distorts labor market behavior in the short run but has a positive impact on (registered) labor earnings in the long run, we should impute a higher MVPF to the program. However, this type of longer-run effect on employment is unlikely, because AFAM did not include any job training or job intermediation component.

An additional value of computing the MVPF is that we can compare our results to those obtained for other similar

programs. Hendren and Sprung-Keyser (2020) report the MVPFs of 133 historical policy changes in the USA. Their evidence suggests lower MVPFs for policies targeting adults/households, as in the case of AFAM, than for policies that directly target children. These lower MVPFs reflect the fact that most of the programs targeting adults reduce (reported) labor earnings through labor market distortions. Thus, as in the case of the AFAM program, the average cost per $1 of mechanical government spending for these policies is greater than $1 (Hendren and Sprung-Keyser, 2020, Figure V). Our MVPF estimate for single-mother AFAM beneficiaries of 0.61 implies an efficiency cost within the range of cash transfer programs targeted to families in the United States, for which Hendren and Sprung-Keyser (2020) estimate an average of 0.74 and a confidence interval of [0.36,1.47] (their Table II). However, our estimate for AFAM is lower than the MVPF of similar programs that induce labor supply disincentives, such as AFDC (MVPF of 0.87). Despite pervasive informality in the AFAM context, the efficiency cost of this policy is still within the range of that of cash transfers in the United States, where informality is substantially lower.

7. Conclusions

The AFAM program's eligibility rules, which are based on verification of reported earnings, create a strong disincentive for registered employment –a notch in beneficiaries' budget constraints. Individuals in households that applied to the program reacted as predicted by economic theory. We built an anatomy of labor market responses induced by the program's financial incentives along four dimensions. First, we established that beneficiaries responded to the program's incentives by reducing their levels of registered employment. Second, we found substantial heterogeneity in these effects: the program had virtually no effect on individuals with a low propensity to be formally employed and a greater-than-average negative effect on individuals with a medium propensity to be a registered employee. Those with a low propensity to work formally did not respond much to the program's financial incentives, probably because they had limited opportunities to work as registered employees to begin with. This suggests that different groups face different incentives and opportunities, and policy design should try to accommodate these differences. Consistent with the findings for developed countries, we also found evidence of strong behavioral responses to social assistance for single mothers. Third, by matching administrative data with a follow-up survey, we provide suggestive evidence that the decline in registered employment translated into both an increase in unregistered employment and a shift towards non-employment. Fourth, in addition to the strong financial incentives of the AFAM program, the behavioral responses we observed translate into an elasticity of participation in registered employment for the whole population of about 0.78, and of 1.3 for single mothers. Finally, the transfers and registered employment responses we estimate yield a MVPF of AFAM on the order of 0.61.

Our evidence shows that neither the extensive margin of (formal) employment elasticity nor the efficiency cost of a welfare program disincentivizing (formal) employment are particularly large in a context of high informality, compared to the context of developed countries such as the United States where informality is much lower. We also highlight important heterogeneity underlying this result. As in developed countries, single mothers are more responsive than is the average beneficiary. Moreover, the elasticity is much greater for workers at the margin of formal employment and much less for (infra-marginal) workers with either low or high attachment to the formal sector, although the efficiency cost does not vary as much between these groups.[45]

Our results allow us to draw valuable conclusions that can be used to minimize programs' disincentives and to improve the equity-efficiency trade-off of policies of this type. In designing cash transfer programs in contexts of high labor informality, policymakers in developing countries must consider reactions along the registered/unregistered employment margin and find ways to mitigate these unintended adverse results, although our findings in this regard are only suggestive and we need more evidence regarding these margins. Mitigating movements from formal to informal employment requires reducing the implicit tax rates or perceived tax wedges on formal earnings, and, wherever possible, modifying the program to incorporate incentives to work formally. Hendren and Sprung-Keyser (2020) show that programs that incorporate these incentives have lower efficiency costs –for instance, the AFDC as compared to the EITC.

A first implication of our analysis is that AFAM authorities (and those designing similar initiatives in countries with widespread informal employment) should consider ways to smooth out the cash notch implied by the program's eligibility rule. For instance, the income test could be modified by introducing a more continuous schedule with phase-in and phase-out regions that withdraw the benefit gradually and with lower implicit tax rates rather than generating a total loss of the benefit at the income threshold. Alternatively, authorities could test the effects of allowing beneficiaries to continue in the program while generating earnings from registered employment above the threshold for a transitory period. This temporary disregard of earnings could ease the transition to formal employment and mitigate the possibility that beneficiaries prefer lower but more stable income from the program to higher but riskier earnings from registered employment. Moreover, the employment of low income workers is often seasonal –e.g., tourism is an important industry in Uruguay with a marked seasonal pattern of high peaks during the three summer months. Allowing short spells of registered employment without total loss of AFAM benefit would probably reduce the negative effects of the program on registered employment.

Our results also highlight the importance of computing the efficiency effects of welfare policies in developing countries. A full welfare analysis might yield unexpected conclusions. For example, in studying the unemployment insurance program in Brazil, Gerard and Gonzaga (forthcoming) found relatively low efficiency costs, in part because high informality levels preclude larger impacts to begin with.

The significant heterogeneity of responses to the program as a function of individuals' propensity to engage in formal employment also has direct consequences for the design of programs of this type, and particularly for their incentive structures. For instance, specific incentives or conditionalities that are costly to monitor (such as minimum working hours requirements) may only be worthwhile for those reasonably expected to react to the program's incentives. The poorest beneficiaries, who are also those with the lowest propensity to be a formal employee—a characteristic that might be proxied by low education levels—seem to be infra-marginal in terms of the formal/informal work decision. This means that concerns about disincentives to pursue registered employment are less relevant for this group. They could benefit more from other policy initiatives such as job training, job intermediation, or public employment programs. As highlighted by Hendren and Sprung-Keyser (2020), the marginal return on policies that transfer resources to adults are relatively low if they are not accompanied by measures to promote beneficiaries' income-generating prospects. At the same time, the high responsiveness of less poor beneficiaries suggests that the program could incorpo-

[45] We thank an anonymous referee for this interpretation of the results.

rate differential incentives for this group. Just as work incentives guided welfare reforms in developed countries, the new generation of cash transfer and social assistance programs in developing countries could incorporate measures to reduce disincentives to engage in formal employment. More broadly, concerns about the efficiency of welfare programs may become more salient as middle-income countries develop and levels of formality increase.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.jpubeco.2020.104313.

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